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Review article

Review on Virtual Power Plants/Virtual Aggregators: Concepts, applications, prospects and operation strategies

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ABSTRACT

A Virtual Power Plant (VPP), Virtual Aggregator (VA), or simply Aggregator, represents the association of several Distributed Energy Resources (DERs) orchestrated to create economic, energy, and social benefits for prosumers, energy markets, and service operators. This review offers a comprehensive analysis of VPPs, covering key topics, including energy markets, required technology, control methods, ancillary services, and financial benefits. Initially, this review discusses the main technologies for implementing VPPs, focusing on energy, informational, and financial exchanges. It categorizes VPPs based on existing literature, both theoretically and mathematically, into Commercial VPP, Technical VPP, VPP 0.0 – 3.0, Fog as VPP, Electrical Vehicle VPP, and Dynamic VPP. Additionally, VPP control techniques are presented, highlighting centralized, decentralized, distributed, and comprehensive controls. The economic aspects are explored, particularly the relationships among new participants in electrical networks, game theory, and tariff characteristics. The review also discusses electricity market models and how VPPs provide ancillary services, analyzing each service in detail. Finally, the global context is considered, emphasizing important cases and their relevance to the reviewed topics. This work aims to highlight the primary aspects of VPPs, providing a solid foundation for their further development and implementation, especially in the context of increasing DER aggregation, by distinguishing the different types of VPPs and their interactions with key aspects of VPP operations.

Contents

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1. Introduction

The global energy landscape is undergoing a significant transformation, characterized by the increasing integration of Distributed Energy Resources (DERs) such as distributed generation (DG), energy storage systems (ESSs), electric vehicles (EVs), and smart loads. This shift towards decentralization marks a pivotal moment in the evolution of electrical networks, as highlighted by the International Energy Agency's report stating that in 2022, the stock of electric vehicles surpassed 10 million and is expected to continue growing rapidly, alongside a projected increase in solar photovoltaic capacity by 1500 GW over the next four years [1,2]. These technological advancements enable individuals, once passive consumers, to become 'prosumers'—active participants in the energy market who produce and make informed decisions about their electricity usage [3]. The move towards decentralization brings twofold benefits and challenges. On the positive side, it enables micro-flexibility from

smaller resources, crucial for integrating non-dispatchable renewable energies that are essential for reducing the carbon footprint [4,5]. However, this shift also imposes significant challenges on traditional electrical networks, which were primarily designed for a centralized, top-down flow of energy. Issues such as line congestion, voltage instabilities, and the unpredictable nature of renewable generation can arise, necessitating innovative solutions to manage these complexities [6].

Acronyms

AI	Artificial Intelligence
AMS	Advanced Microgrid Solutions
AS	Ancillary Services
BESS	Battery Energy Storage System
BRP	Balance Responsible Party
CCC	Control Coordination Center
CPP	Critical Peak Pricing
DAM	Day Ahead Market
DERMS	Distributed Energy Resource Management Systems
DG	Distributed Generation
DSM	Demand-Side Management
DSO	Distribution System Operator
DVPP	Dynamic Virtual Power Plant
ESS	Energy Storage System
EV	Electric Vehicle
EV-VPP	Electrical Vehicle Virtual Power Plant
FiT	Feed-in Tariff
GMP	Green Mountain Power
LP	Linear Programming
MILP	Mixed Integer Linear Programming
NEM	National Electricity Market
OPF	Optimal Power Flow
PHEVs	Plug-in Hybrid Electric Vehicles
PV	Photovoltaic
RES	Renewable Energy Sources
RL	Reinforcement Learning
RTP	Real-time Pricing
R-VPP	Retailer Virtual Power Plant or Virtual Power Plant 0.0
ToU	Time of Use
TS	Transmission Systems
TSO	Transmission System Operator
V2B	Vehicle-to-Building
V2B2	Building-to-Vehicle-to-Building
V2G	Vehicle-to-Grid
V2H	Vehicle-to-Home
V2L	Vehicle-to-Load
V2V	Vehicle-to-Vehicle
V2X	Vehicle-to-Everything
V4G	Vehicle for Grid
VGI	Vehicle-Grid Integration
VPP	Virtual Power Plant
VPP 1.0	Virtual Power Plant 1.0 or System-focused Virtual Power Plant
VPP 2.0	Virtual Power Plant 2.0 or Consumer-focused Virtual Power Plant

Variables

$x_{a,t}^{\text{net}}$	Net electricity required by user a during time-slot t.
$x_{0,t}$	Total demand during time-slot t.
$C(x_{0,t})$	Aggregated cost function of the total demand at time t.
$\sigma_t^{ft/tou}$	Flat/ToU tariff applied to the energy imported from the grid at time t.
σ_t^{fit}	Feed-in tariff applied to the energy exported to the grid at time t.
$p_{i,t}^g$	Active power generated at node i at time t.
$q_{i,t}^g$	Reactive power generated at node i at time t.
$v_{i,t}$	Voltage at node i at time t.
λ_t	Price update variable associated with the power balance constraint at time t.
$L_{\text{VPP}}(x,\lambda)$	Lagrangian function of the VPP optimization problem.
$D(\lambda)$	Lagrange dual function associated with the VPP optimization problem.

To solve these issues, some possible approaches emerge to integrate a large number of DERs into power systems as explained in [7]. One possible solution, for example, is the Local Energy Markets (LEMs). It corresponds to a form of organization in which buyers and sellers trade energy directly with each other within a specific geographic area. In the electricity sector, this market is designed to facilitate the negotiation of small-scale energy, usually, within a community [8]. It is particularly interesting for local prosumers, as the LEMs system stimulates and benefits them by generating a more decentralized, democratic, flexible, and resilient energy model for negotiations. Although various technologies are used to maintain balance between supply and demand and to favor communication between the parties [9], normally LEMs are mostly facilitated by a digital platform, which provides a place for energy negotiations. In recent years, this concept is gaining popularity in the energy sector as it offers the potential to reduce energy costs, promote renewable sources, and increase energy security and resilience by creating a more distributed energy grid and a bigger energy flow distribution [10].

Another concept that has attracted attention in scientific circles is the Virtual Power Plant (VPP). It represents the association of multiple DERs that function as a single intelligent entity [11] and brings together different flexible energy resources, facilitating the decentralization of the grid and allowing their participation in the market as a unified group. Moreover, this organizational model offers the advantage of allowing participants to operate remotely without requiring physical proximity. This increases the diversity of conditions among members, which in turn strengthens the group.

Moreover, VPPs offer a pathway to enhance energy security and resilience by enabling more efficient and flexible energy distribution. By aggregating the capabilities of diverse DERs, VPPs can respond dynamically to fluctuations in energy supply and demand, mitigating the risks associated with the intermittent nature of renewable energy sources. This adaptability not only supports the stability of the grid but also promotes the increased adoption of renewable energy technologies, aligning with global sustainability goals.

Furthermore, the role of VPPs in enhancing energy security and resilience is particularly significant in the context of growing renewable energy integration. By leveraging advanced analytics and real-time data, VPPs can optimize the operation of diverse DERs, ensuring a balanced and stable energy supply even during periods of high demand or low generation from renewable sources. This capability to dynamically adjust to energy fluctuations not only mitigates the risks associated with the variability of solar and wind power but also reduces reliance on traditional fossil-fuel-based generation. As a result, VPPs contribute to a more sustainable energy landscape by facilitating the seamless incorporation of renewable energy into the grid. However, realizing the full potential of VPPs requires overcoming several challenges, including the need for robust regulatory frameworks that support decentralized energy management, the development of standardized communication protocols for interoperability, and the creation of market structures that incentivize the participation of small-scale energy producers. Addressing these challenges is essential to unlock the benefits of VPPs, ultimately fostering a more resilient, efficient, and sustainable energy system for the future.

In addition to the technical and operational advantages, VPPs also offer significant economic and environmental benefits. Economically, VPPs can lower energy costs for consumers by optimizing the use of distributed energy resources and reducing the need for expensive peak generation and grid upgrades. They enable better utilization of existing infrastructure and can defer or eliminate the need for new investments in transmission and distribution systems. Environmentally, VPPs support the integration of renewable energy sources, thus reducing greenhouse gas emissions and reliance on fossil fuels. By promoting a decentralized and diversified energy portfolio, VPPs enhance the overall resilience of the energy system to climate-related disruptions. This approach aligns with global initiatives to combat climate change and transition to sustainable energy systems. However, for VPPs to realize these benefits fully, there must be continued innovation in energy management technologies and supportive policy frameworks that encourage investment and participation from a broad range of stakeholders, including small-scale prosumers, utility companies, and regulatory bodies.

The first research published about this type of association was from Dr. Shimon Awerbuch in 1997, at that time, he used the term "Virtual utility" to reference a new form of flexible collaboration between utilities to provide electrical services to customers by sharing the results of their assets [12]. Over the years, this concept and the technologies involved have improved a lot, reaching the state that currently exists. Despite this evolution, there is a lack in the literature of a clear and comprehensive review of the various aspects and possibilities presented by VPPs. This work aims to accomplish this feat.

1.1. Literature meta-review

Given the broad scope of the present work and its inherent review characteristic, we reserve this particular section to discuss the origins of the VPP concept stemming from smart grids and microgrids, as well as providing a meta-review of other VPP review works. This also allows to pinpoint literature gaps and highlight the contributions thereafter.

1.1.1. Smart grids

Since their inception, electrical systems around the world have undergone a few structural changes in their operation. However, from the 2000s onward, decentralization, automation, and the use of highly sophisticated technologies began to significantly change the way relationships in the electrical grid operate [13]. This paradigm shift generated the concept known as Smart Grids (SGs). It involves a bidirectional flow of data and energy in a network that is integrated with communication, measurement, and control technologies. Although its origins are not entirely clear, one of the earliest pioneers was the Italian project Telegestore, which interconnects approximately 32 million households using smart meters [14].

In recent years, studies and applications related to this concept have grown exponentially and, as a result, this theme has several branches. Some more comprehensive articles about this topic are [14–16]. The authors present fundamental technologies for the implementation of SGs and the context of several countries in [14]. On [15], the focus is not only on the technological aspect but also on a series of challenges faced in these new grids. Finally, authors in [16] provide a broad review of the development of smart grids, its main challenges, and perspectives. Additionally, it also focused on various topics such as distributed generation, microgrids, smart meters' deployment, energy storage technologies, and the role of smart loads in primary frequency response provision. The main features of these and other articles in the same pattern are not only the clarity with which the ideas are presented but also the attention given to the history, development, functionalities, technologies, challenges, and perspectives, giving a complete overview of SGs.

Although these comprehensive models of the subject are important, they have gradually been replaced by more specialized ones as the theme has been further developed. Because of this, one of the foci of the lines of research is in the area of communication and information and one of the most comprehensive reviews in this area is [17], which is notable for its development of advanced concepts related to technologies that have enabled the growth of SGs, including the architecture of cyber–physical systems, cybersecurity, cloud computing, Big Data, the Internet of Things, and the Internet of Services. It also draws attention to the association of intelligent networks with Industry 4.0 (SGI 4.0), which was also addressed by several other works [18–20]. Other recent texts have also developed subjects related to these, such as [21], which presented some of the previously mentioned concepts applied to smart grids, and [22], which developed the cloud concept associated with smart metering to enable the effective integration of various resources in low-voltage distribution networks. Additionally, the book [23] discusses modern methodologies and technologies related to power electronics, signal processing, and communication systems applications in SGs.

Other significant points involving this new way of seeing the flow of electric energy are the issues of control and orchestration. They warrant attention primarily because these new networks have a high penetration of renewable energies. At first, the positive characteristics of this association are esteemed/ however, if not properly managed, problems such as energy congestion or voltage fluctuations can occur [24]. In [25], various stochastic optimization methods for smart grids are presented. Authors in [26] also bring a complete review of optimization methods, integration with RES, and challenges. The concept of Advanced Distribution Management (ADM) also deserves to be highlighted because it is related to this energy transition [27]. It corresponds to a specific set of technologies and processes used to monitor and control the distribution network in real-time. ADM systems use data from sensors and other network devices to analyze the state of the network and manage its operation with goals including optimizing the use of distribution assets, improving power quality, and managing power outages. Although some authors treat SGs and ADM as distinct concepts, it is clear that they overlap in many aspects.

Another key aspect involving SGs is the issue of demand-side management. This corresponds to the planning, implementation, and monitoring of pre-defined activities that affect consumers of electricity. In [28] the integration of EVs is analyzed from the flow of energy to the grid to the provision of ancillary services which represents a complex union due to the random character of mobility presented by EVs. Along with that, another subject that gained prominence in networks is different advanced control methods [29,30] like the Model Predictive Control (MPC), which is an advanced control processing method that predicts future outputs based on historical data [31].

1.1.2. Microgrids

This growth of smart grids has given rise to new concepts and applications, including Microgrids. A microgrid is a localized electricity distribution and consumption network that can operate independently, ensuring a reliable power supply in the area it serves [32,33]. It typically incorporates a mix of renewable energy sources like solar panels and wind turbines, as well as energy storage technologies such as batteries or fuel cells. Microgrids offer several benefits, such as keeping systems stable during network disruptions, providing ancillary services, encouraging the use of clean energy, etc.

Several recent studies have explored technical, economic, organizational, and technological aspects of microgrids, such as [34–36]. However, most of the papers tend to focus on more specific subdivisions of microgrids. In [33], for instance, a complete review of control and stability techniques for different types of DC, AC, and hybrid (DC/AC) microgrids is developed and special attention was given to the methods of the interconnection of AC and DC microgrids in a hybrid AC/DC microgrids. Similarly, the review [37] also brought dynamic modeling, stability, and control ideas to interconnected microgrids (IMGs).

Microgrids also stand out for the high presence of renewable sources. In [38], authors conducted a series of bibliographical analyses and concluded that the fields of microgrids, renewables, and optimization hold tremendous promise. An example of this is the significant increase in keywords related to these concepts between two analyzed periods (2005–2011 and 2016–2021), which rose from 138 to 699, representing a 406% growth. In [39] a series of economic and environmental advantages are demonstrated through studies and simulations, demonstrating a set of possible advantages due to the insertion of microgrids in the network. Ref. [40] explored a combined heat and power (CHP) microgrid model with renewable sources, optimizing results in the island operation. And studies based on zero microgrids dependent on carbon sources are already being developed as in [41–44], which mainly studied the feasibility of associating this model with hydrogen sources.

Although their implementations are currently increasing, microgrids bring new challenges to electrical networks. In [45] a complete study on these new challenges is provided. The article draws several significant conclusions. Firstly, it emphasizes the crucial role of prosumers in the context of direct low-voltage current (LVDC). It also highlights the importance of microgrids in promoting social welfare. Additionally, the article underscores the facilitative nature of bus distribution in integrating renewable sources into the grid. Furthermore, it points out the challenges associated with the lack of standardized regulations, particularly in terms of voltage selection. Lastly, the mass adoption of prosumers can potentially give rise to operational issues.

The concept of Virtual Power Plants came up to correct some of these and many other issues. As previously mentioned, these models of association aim to bring together a series of DERs, which include microgrids, to achieve economic, social, environmental, and energy benefits to society. This review intends to explain and detail all these aspects, but first, an analysis of existing papers will be presented.

1.1.3. Virtual Power Plants

As it is a recent topic, VPPs do not have a single scientific standard. In reality, what is perceived after researching the most varied articles are attempts to understand and improve the various branches of VPPs. Still, publications often follow their paths through the development of new categories, concepts, and logic, making the subject richer and more complex, which is normally expected from relatively new themes in science.

In trying to bring together these existing pieces of knowledge about VPPs, several reviews have started to appear in the last few years. Some of them that serve as an introduction for their range of topics and clarity are [46–51]. They, however, lack overall technical depth and end up being shallow on many topics.

In parallel, [52] not only provides a broad overview of VPPs but also thoroughly describes key concepts in implementing these groups. It stands out mainly in the analysis of bottom-up operational schemes and their application in Net Zero Energy. The examples brought are also current and the definition, synthesized after a long investigation of several articles, represents well what VPPs are in fact. In [53] several aspects are emphasized, but a more specific and detailed look is given to themes that the previous article did not deal so much with. Among them, the part of Internal control of VPP, Service-Oriented Optimization Algorithms, and Electricity Markets is very well-developed. In [54] broad aspects are also developed, even though the main focus is the transformation of microgrids into VPPs. Although these works bring a wide view of concepts, they often fail to delve into certain more specific topics. For this reason, other types of papers arise intending to analyze these particular subjects. A commonly followed path for these studies is the economic one.

In [55] there was research on different models of VPPs interacting in various environments of electricity markets. Through a series of models, optimization techniques, and case studies, the authors exposed some recent, realistic, and profitable schemes for the sets. Another unique article that was also very well-structured and developed is [56]. It investigates the possibility of implementing a stochastic and renewable VPP in the European Power Exchange/European Energy Exchange. After analyzing a series of models involving photovoltaic, biogas, and battery systems, the authors concluded that the marginal prices of each component and the availability of energy are crucial for the VPP to have an optimal economic operation. It is important to highlight that the time of year greatly interfered with the biogas/photovoltaic energy ratio in the model simulations, which affects the overall performance, with the greatest variations in winter and summer. In [57] the optimal operation of VPPs is given by focusing on the investor's perspective. Its uniqueness lies in the use of Markowitz portfolio theory and genetic algorithms in developing the configuration of DER capabilities. The internal division of the VPP into classes and the use of thermal energy to avoid supply fluctuations are also interesting ideas.

In [58] other economic ideas are developed, focusing on the development of different business models associated with VPPs. Concerning this theme, the book [59] is outstanding. It gives a comprehensive study of various VPPs' business models and presents reliable operation strategies under different scenarios. The applied models have great relevance in the optimal functioning of the set and arrangement of the DERs, being an important guide in the integration of the VPPs. The book also covers demand-response business operations, ancillary services markets, dynamic pricing strategy, and an analysis of reliable integrated power system operation with VPPs.

Another important branch for the implementation of these groups is scheduling problems. Authors of the review manuscript [60] effectively covered most of the aspects associated with DERs in microgrids and VPPs. It brings a series of techniques, models, and points of view on these issues and synthesized them completely and uniquely, valuing above all uncertainty, reliability, reactive power, control, emission, stability, and demand response, synthesizing different articles on the subject. In addition, it also draws attention to little-researched points in VPPs and microgrids. First, there are few studies associating reliability considerations and emission problems with scheduling, while about second, DERs such as pump-storage and hydro-power, stability, and demand response considerations do not have the attention they should. Another interesting review of energy management for VPPs is done in [61], which not only brings many of the concepts seen in the previous one but also some new features such as scheduling programs taking into account an analysis of the joint technical and economic constraints considering different network levels.

A topic that lacks in-depth analysis is the technology involved in the process of implementing VPPs. A major exception to this situation is the paper [62]. Unlike others, it stands out for going far beyond just presenting the necessary basic components. It breaks down the implementation of VPPs into a series of layers: Components, Communication, Data, Management, and Framework, describing not only the usual technologies but also what is to come in the next generations of smart grids. It focuses on a series of important factors for the correct implementation like cybersecurity, blockchain, cloud computing, edge computing, machine learning, artificial intelligence, etc. By the end, the paper gives an important analysis of the current scenario and what is expected in the future, highlighting important points about the relationship of microgrids and VPPs with new components. It is also possible to find other more basic reviews on this subject such as [55].

Additionally, it is worth noting the similarities between Virtual Power Plants (VPPs) and Distributed Energy Resource Management Systems (DERMS), as they aim to accomplish nearly the same objectives: both systems are used to manage and coordinate with DERS and integrate with grid operations and market systems to provide grid services. However, DERMS are utility Enterprise solutions that are owned by the utility [63].

1.1.4. Comparisons between local energy markets, peer-to-peer and Virtual Power Plants

As seen, the vast majority of countries are looking for renewable sources to compose their energy matrix, but this integration brings a series of challenges for grid management due to its stochastic nature. Therefore, the continuous and reliable supply becomes complex and unpredictable. Given this context, DERs bring a series of facilities for the integration of renewable sources, mainly because they are adaptable according to the individual conditions of consumers and climatic conditions. In addition, due to their spatial distribution, they bring a set of advantages when compared to larger plants, which are typically located far from consumer centers, while DERs can supply energy directly to customers, reducing the burden on the transmission and distribution network.

Different ways of coordinating the operation of DERs have emerged over time, with some of the most prominent currently being the concepts of VPPs, Peer-to-Peer (P2P), and LEMs [64,65]. All of these methods are related, to some extent to the direct optimization of prosumers [66], and to the decentralization and modernization of energy systems. As such, electricity is consumed, produced, and distributed with a more local emphasis [67,68]. As previously discussed, VPPs connect different DERs to allow for a more diversified energy generation portfolio and improve the grid's overall efficiency. P2P, on the other hand, allows small individual producers and companies to exchange energy among themselves, bypassing – or rather, complimenting – the concept and need of the traditional energy retailer. LEMs are a generalization of methods such as P2P, in which any LEM emphasizes the analysis of local energy exchanges between producers and consumers spatially close to each other, presenting a more formalized market through regulations [69].

As will be seen, the P2P system can be incorporated internally by the VPP, presenting itself as a decentralized form with less control. Regarding foreign trade, VPPs can participate in a P2P system as an energy supplier. Therefore, as means of organizing DERs, the concepts of VPPs and P2P overlap and are related in many ways. However, the former has several advantages over the latter [70], being more tangible and less idealistic in practice. VPPs in general are more reliable to unite the different resources under the direct or indirect control of an entity, including more decentralized approaches, while P2P methods rely on the decisions of individual households and companies to manage their DERs, which can lead to a lower reliability across the network [71]. In addition, because VPPs are one entity controlling several resources, they can present much greater flexibility in controlling loads and generators, while this does not apply to the immense decentralization of P2P. VPPs are also much more energy efficient, as this is one of the main foci of their implementation, which is the result of a series of optimization analyses, spatial distribution, weather conditions, storage systems, etc. P2P, on the other hand, are controlled by individuals and companies, who often do not care about such points or even do not have enough knowledge of electricity to manage and optimize their production and expenses [7,72]. Also, as will be seen throughout the text, the organization of VPPs can bring a series of benefits to the electrical system through ancillary services, which is not easily applicable in P2P methods.

Furthermore, in P2P settings, prosumers and consumers interact directly, which can further marginalize the role of Distribution System Operators (DSOs), since they are heavily focused on the financial advantages for prosumers. It is also worth noting that such shift towards prosumer-centric financial advantages is one of the hurdles in large-scale adoption of P2P, since the distribution system operators and retailers have little incentive to foster P2P systems within their own networks [73]. An example of a P2P structure that is having difficulties in its implementation due to all the factors mentioned above is the Power Ledger project in Australia [69]. It is a project founded in 2016 that aims to create a decentralized electricity market. As it is based on P2P energy trading, consumers and prosumers can exchange energy in real-time through an online platform. All exchanges would have their security guaranteed by the blockchain system, which will be developed later in this review.

With such analyses and examples, it can be seen that P2P is a much more difficult system to implement. LEMs in general also face some of these difficulties, in particular on decentralized approaches; however, as they have a more generic structure than P2P methods, their organization is prone to more diversified trading approaches, in which problems can be attenuated and even overcome. Some papers that provided a good analyses on LEMs are provided for further reading in case the reader might be interested [73,74]. It is also important to emphasize that VPPs and LEMs can complement each other in specific contexts. By representing a system that stimulates the production of energy in a decentralized way in certain regions, LEMs provide a marketplace where VPPs can grow and negotiate electricity. On the other hand, VPPs strengthen the existence of LEMs, as it optimizes the distributed resources of the specific region and makes negotiation participants stronger, more resilient, and more productive, as one of the main consequences of the enrichment in market negotiations.

This type of analysis makes it clear that VPPs bring the greatest benefits to all participants because it is a more tangible and less idealistic application. Precisely for this reason, this concept has shown exponential growth in recent years, which leads to the aggregation of its main information extremely important for the area.

1.2. Literature gap

Numerous research studies have reviewed the VPP from different perspectives, which is rather involving around the operation, structure, etc. For an energy system that aggregates different units together, several important aspects are not explored.

First, the ancillary services provided by the VPP are not elaborated. The different potential ancillary services are one of the most important characteristics of VPP. Despite some of the texts seen having a subsection for this topic, none of them deepened into it. Second, it is a lack of description of the real interaction of these new participants in the electricity ecosystem. A study on the importance of the tariff system in the implementation and permanence of VPPs and DERs, in general, is also missing. In addition, the development of concepts is also limited especially in the broader reviews. With the continuous development of VPP, different evolution of VPP should be reviewed comprehensively in combination with the conventional CVPP and TVPP. Last but not least, although many papers mention VPPs around the world, few of them explicitly relate these aggregates to the regulatory and contextual situations of each country or continent, which make concept development seem unrealistic in practice. Therefore, in this paper, we give a more comprehensive review of VPP in different aspects mentioned above, providing a reference for the development of VPP in the long run.

1.3. Contributions

The construction of this article focuses on bringing the most diverse aspects of VPPs clearly and comprehensively. It stands out from the rest for several reasons, besides the comprehensive meta-review of other smart grids, microgrids and VPP review papers presented previously.

The first contribution is a description of various VPP concepts worked on by different academic circles. Although many of them overlap, each is thoroughly detailed theoretically. Another unique feature is the description of new concepts with mathematical equations and deductions, which no other review has done. All of them are also related by a diagram, that clearly explains connections between different classes and their definitions, demonstrating the complexity of the theme. All of this can be found in Section 2.

The second contribution lies in the analysis of articles that made a critical and realistic investigation of the entry of VPPs in the electricity markets. In general, the review papers are limited to considering only the positive sides of this entry, but new participants in the electricity ecosystem of the countries will generate imbalances and internal difficulties, which also need to be carefully studied and referenced. Furthermore, the examination of papers focused on the tariff issues within the VPP is a differential as well. Despite being a little-researched topic, it presents some important studies that were neglected by previous research. This is done throughout the entire text, and in particular within Sections 5 and 6.

The third contribution is a state-of-the-art analysis of ancillary services related to VPPs and DERs. Unlike other reviews, we perform a comprehensive examination of crucial services such as frequency regulation, voltage regulation, reactive power, scheduling and dispatch, among others. By emphasizing the importance of these ancillary services in VPPs, the present research sheds light on new capabilities and benefits of these new organizations of DERs in power grids. While this is a rich application feature, few works have explored the various types of existing services from the perspective of VPPs beyond frequency regulation and, in particular, their potential symbiosis. These analyses can be found throughout the text, but their main focus is on Section 6.7.

The fourth contribution consists of a descriptive table showing various cases of VPPs around the world, providing a comprehensive study of the current context in different locations. The table includes examples from different continents, which are correlated with the VPP categories described in the previous sections. In addition, we explain the importance of adopting regulations consistent with the energy context of the countries for the development of distributed resources, thus reinforcing the realistic examination of the implementation of these technological sets. Notably, we highlight recent cases in Denmark with Market Model 3.0 and in Germany with Redespatch 3.0, which serve as illustrations of regulatory updates that are adapting to changes in the electrical grid. These studies begin in Section 7, and the last 2 examples are in Sections 7.5.1 and 7.5.2, respectively.

The fifth major contribution of this manuscript is to establish a clear correlation between various VPPs categories and their implications in economic modeling and ancillary service provision. This examination is carried out systematically across different sections of the text, however, it is better detailed in Sections 5.5 and 6.7.5. In addition, this review also highlights the distinctions between the implementation of LEMs, P2P and VPPs in several sections, such as the ones mentioned above. In doing so, it also sheds light on the inherent limitations of the P2P model when compared to VPPs, highlighting the challenges associated with its zero-net game nature during implementation.

The outcome of this review is projected to serve as a foundational reference for future research and development in the field of Virtual Power Plants (VPPs). By systematically examining and categorizing various VPP concepts, this work provides a detailed and nuanced understanding that can guide both academic research and practical implementations. The comprehensive meta-review offers insights into the integration of DERs, the operational dynamics within electricity markets, and the provision of ancillary services, establishing a robust framework for evaluating the economic and technical impacts of VPPs. Additionally, the critical analysis of tariff issues and market participation challenges presents a balanced view of the potential barriers and opportunities in the deployment of VPPs. This review aims to foster further innovation by highlighting the synergies between different technological advancements and regulatory developments, paving the way for more resilient, efficient, and sustainable energy systems. By addressing the identified gaps and presenting new perspectives, the present work not only contributes to the existing body of knowledge but also sets the stage for ongoing advancements in the decentralization and democratization of energy networks.

It is important to mention that an aggregator may be understood as the consolidation of various agents within a power system, including consumers, producers, prosumers, or any combination thereof. This unified entity operates in power system markets, participating in both wholesale and retail transactions, as well as providing services to the operator. Essentially, an aggregator is a company that manages a VPP, which in turn is a collective assembly of DERs [75]. While some authors distinguish between VPPs and aggregators, this article, along with many others, uses these terms interchangeably. Furthermore, the concept of aggregators can also be seen as an evolution of VPPs, as the VPPs were initially associated with high-voltage networks, while aggregators encompass both low and high-voltage networks, providing a more comprehensive solution for the current electrical grid. However, as said before, even considering these differences, the main objective and all the concepts behind the organization of these sets of DERs are extremely similar, which justifies the use of terms as synonyms in this context.

The rest of the review was organized following the paths explained in Fig. 1:

Section 3 analyzes the different components present in these aggregates highlighting the simplest, from DG, ESSs, EVs and controllable loads, to more advanced technologies such as Information and Communication Technology (ICT), Intelligent control, Edge computing, Smart inverters, advanced communication protocols, Blockchain and Peer-to-Peer energy trading. Section 2 provides a series of classifications of VPPs, highlighting not only the CVPP/TVPP but many other classes. Section 4 studied organizational control and optimal energy management to orchestrate DERs harmoniously and functionally to achieve the objectives of the group. Among them, 4 categories stand out nowadays: Centralized Control, Decentralized Control, Distributed Control, and Comprehensive Control. Section 5 brought economic analyses of VPPs. It covered a range from game theory, tariffs, and economic analyses of different categories to price forecasts for the technologies involved. In Section 6, the operation of VPPs in electricity markets, and ancillary services were detailed, focusing on frequency, reactive power, voltage control, scheduling and dispatch, and operational reservations. Lastly, Section 7 brought a set of cases around the world relating them to the concepts presented throughout this review. An analysis was also made of the evolution of laws involving electrical networks, their future perspectives, and how these legislative changes can significantly affect the development of VPPs and the democratization of energy in different countries.

2. Virtual Power Plants definitions and categories

Virtual Power Plants (VPPs) can be categorized according to different criteria. Fig. 2 illustrates the primary classifications and their interrelations.

2.1. Commercial VPP and technical VPP

There are two main recurring categories in the texts: Commercial Virtual Power Plant and Technical Virtual Power Plant [54]. They are more generic and can cover all cases.

The first represents the implementation with a main focus on the financial and economic benefits produced by the combination of DERs, which encompass both the individual participants and the whole [76]. It can also provide important services like peak shaving, valley-filling, etc. [77]. As the primary objective of CVPP is economic optimization [78], financial parameters such as cost, revenue, Net Present Value (NPV), Payback Period, and profits are crucial factors in determining its implementation. They are usually owned and operated by commercial entities, and for this category to work, it is important to plan the relationship between the participants very well to avoid unnecessary energy and economic losses by contracts.

The second definition is presented as the implementation of the VPP considering the operational constraints of the network and focusing on social welfare by providing the energy of quality [79]. To achieve this, the following are used: complex optimization calculations, efficient use of energy, applications of the best accessible technologies, monitoring,

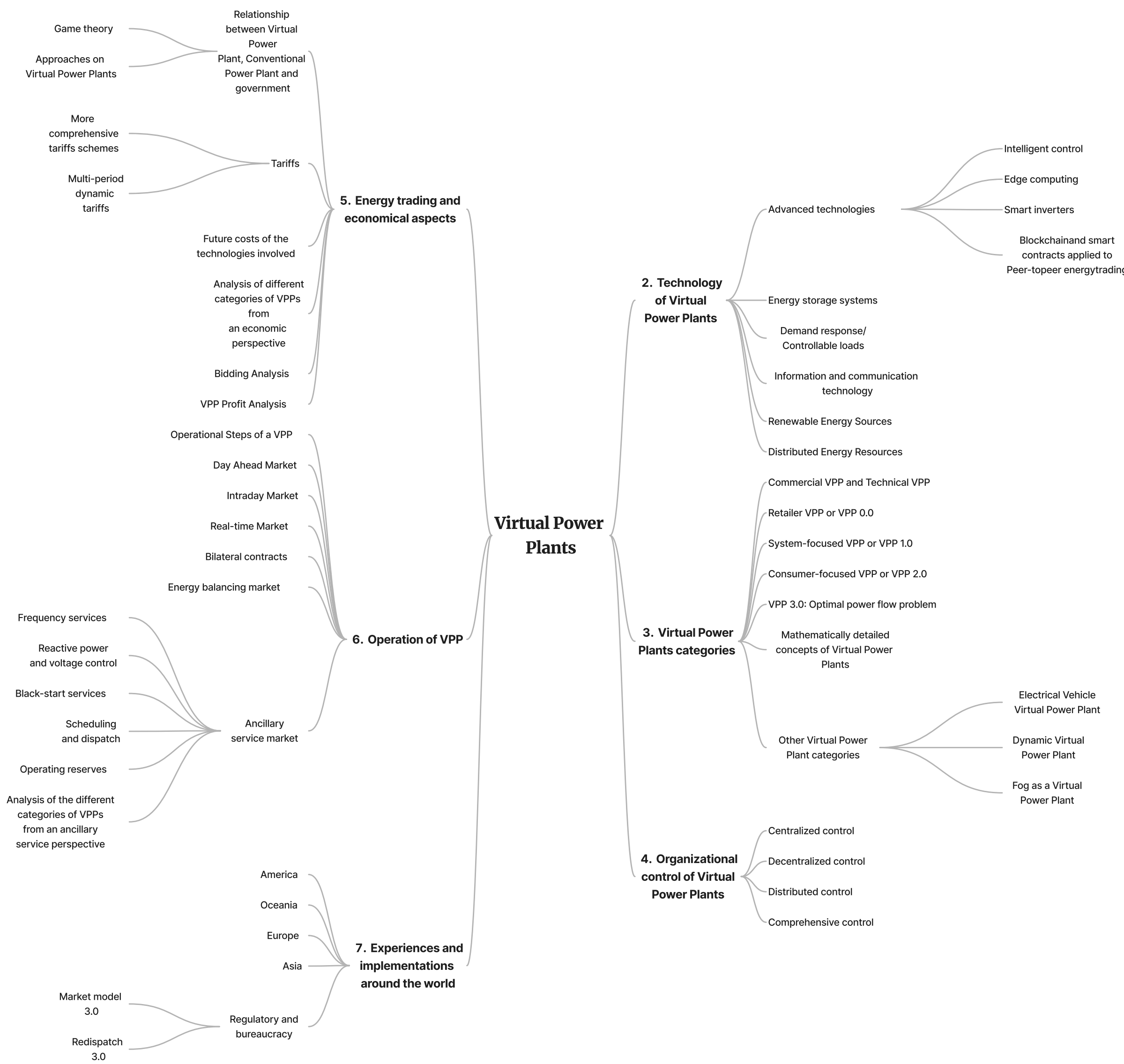

Fig. 1. Theoretical organization of the review.

and data forecasting. In addition, it is important to mention the contribution of TVPP in the resilience and operation of the network [80] by providing ancillary services, highlighting the regulation of voltage, frequency, in black-start services and the balance of energy supply and demand. Here, economic issues are placed as a secondary objective, even though, many times, there are great financial benefits in the use of TVPP too [81]. It is important to emphasize that, as this category is often owned and operated by utilities or grid operators focusing on technical aspects [82], it is more susceptible to receiving subsidies and grants from the government or from companies interested in benefiting from the knowledge attained.

2.2. Mathematically detailed concepts of Virtual Power Plants

Virtual Power Plants (VPPs) are a new approach to electricity generation and distribution that is becoming increasingly important. By aggregating distributed energy resources, VPPs can provide more flexibility and resilience to the power grid, however, their reliance on renewable energy sources can create challenges for maintaining the stability of the system. Although energy storage systems can help to manage system fluctuations, they are not sufficient on their own to ensure the optimal functioning of VPPs. Accurate prediction of energy production and consumption is crucial for maintaining a balance between demand and production, and AI-based algorithms are emerging as a popular solution for this purpose. Additionally, precise weather prediction is essential, and specialized software algorithms are necessary to achieve this. To keep up with changes in the power system, optimization, and forecasting algorithms should be updated regularly.

Achieving optimal functionality in VPPs and electrical networks requires a comprehensive approach involving Programming Models, Optimization Problems, Multiobjective Optimization Algorithms, and Distributed Optimization Algorithms. These methods are designed to address a range of constraints, and within each category, there are numerous sub-methods with unique advantages and disadvantages. By carefully considering and applying these approaches, VPPs can be operated correctly. Some articles that have already carried out good analyses on these more specific topics are [53,60].

This section of the review will emphasize more specific and mathematically detailed concepts than the previous one. Three main practical models will be presented and, later, related to the problem of optimal power flow. It is noteworthy that the models presented bring more tangible applications and methodologies, however, as the VPP is constantly evolving, they can be seen more generically and broadly over time. These concepts were mentioned more timidly in the following

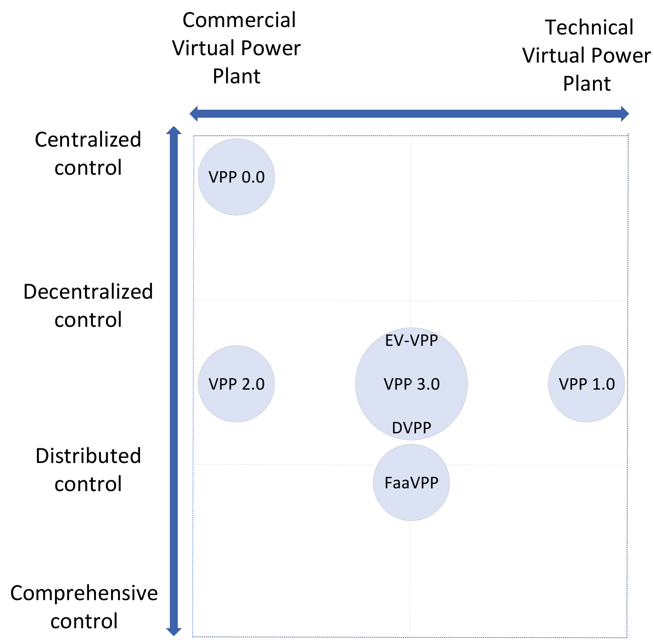

Fig. 2. Categories and relationships among VPPs. On the horizontal axis are the two most generic classifications of VPPs: CVPP and TVPP, the first being related to economic parameters and the second to technical ones, more information can be found in Section 2.1. The different control models are organized on the vertical axis, and their respective descriptions and analyses can be found in Section 4.

articles [83,84] and more explicitly in [7,85,86]. They represent equations and descriptions taking into account a series of realistic network constraints.

2.2.1. Retailer VPP or VPP 0.0

The first category to be explained is the Retailer VPP (R-VPP) Model or VPP 0.0. This model falls within the CVPP category, as its primary goal is to secure better prices and rates for VPP members, thereby generating individual savings. To achieve this objective, DER owners must relinquish some decision-making at specific times, allowing a central control system to make those decisions on their behalf. Unlike other models, the R-VPP does not utilize optimization but strategically places DERs in different parts of the Medium Voltage network to mitigate price exposure, as shown in Fig. 3.

Despite its simplicity, the R-VPP model has financial benefits. It is not yet widely implemented due to the initial investment required for technology, but as costs decrease and research progresses, the R-VPP is expected to become more common. The model can be mathematically described by the following equation:

$$\begin{aligned} &\underset{x_0 \in \mathcal{X}_0}{\text{minimize}} \; f(x_0, T) \\ &\text{subject to} \sum_{a \in A} x^{\text{net}}_{a,t} = x_{0,t} \quad \text{for } t \in T \end{aligned} \tag{1}$$

where $x^{\text{net}}_{a,t}$ represents the net electricity required by user α during a time-slot. This optimization problem aims to minimize the function $f(x_0, T)$ with respect to the variable x_0, subject to the constraint that the sum of the net flows of all nodes a in the set A at each time t equals $x_{0,t}$.

Practical Example:

A VPP 0.0 consists of 50 small solar energy producers, each generating an average of 100 kWh per month. This energy is sold to the grid at a feed-in tariff of 0.04 €/kWh. The VPP negotiates with the energy supplier to purchase electricity at 0.10 €/kWh, with a standard consumption tariff of 0.15 €/kWh.

The cost function for a producer A who consumes $x_{A,\text{consumed}}$ kWh and produces $x_{A,\text{produced}}$ kWh is given by:

$$g_A(x_A) = (x_{A,\text{consumed}} \times \text{purchase price}) - (x_{A,\text{produced}} \times \text{feed-in tariff})$$

For a producer who consumes 120 kWh in the same month, the cost function is calculated as:

$$g_A(x_A) = (120 \text{ kWh} \times 0.10 \text{ €/kWh}) - (100 \text{ kWh} \times 0.04 \text{ €/kWh})$$

$$g_A(x_A) = 12.00 \text{ €} - 4.00 \text{ €} = 8.00 \text{ €}$$

In this scenario, the VPP 0.0 helps reduce energy costs for participating producers by centralizing the negotiation of tariffs. The optimal result is obtained by minimizing the total net cost for each producer, ensuring that the purchase of energy from the grid and the sale of generated energy are done at the best possible rates.

Practical Example without VPP:

Consider the same scenario without the VPP, where individual producers directly interact with the grid without centralized control. Assume the cost of energy purchased from the grid is higher due to lack of negotiation power and the inability to collectively bargain for better tariffs.

For a producer who consumes 120 kWh in the same month and sells 100 kWh to the grid, the cost function without VPP might be:

$$g_{A,\text{noVPP}}(x_A) = (120 \text{ kWh} \times 0.15 \text{ €/kWh}) - (100 \text{ kWh} \times 0.04 \text{ €/kWh})$$

$$g_{A,\text{noVPP}}(x_A) = 18.00 \text{ €} - 4.00 \text{ €} = 14.00 \text{ €}$$

In this simple example is possible to see that implementing the VPP 0.0 results in a cost reduction highlighting the economic benefits of the VPP model. Additionally, the VPP 0.0 can negotiate better tariffs with energy suppliers, ensuring lower purchase prices and better rates for energy sold back to the grid. Producers also benefit from simplified energy management and decision-making processes, as the central control system optimizes energy flows.

2.2.2. System-focused VPP or VPP 1.0

The system-focused VPP, or VPP 1.0, is concerned with the maximization of social welfare. It is part of the TVPP category, as it focuses on improving the operation of the network system and is well-suited to providing ancillary services. However, it still participates in the wholesale market and generates advantages for its members. Therefore, the main objective of this VPP is similar to that of the TVPP, aiming to function close to the ideal in terms of energy efficiency while considering the context and limits of the entire system. The cost function for the VPP can be described as follows:

$$\begin{aligned} &f(x_0) = \sum_{t \in T} C(x_{0,t}) \\ &\text{where } \; C(x_{0,t}) = c_2 (x_{0,t})^2 + c_1 x_{0,t} + c_0 \end{aligned} \tag{2}$$

where $x_{0,t}$ is the total demand during time t and C denotes the aggregation of the cost. c_0, c_1, and c_2 are parameters related to the price variations in the wholesale market.

This problem can be transformed into an optimization problem focusing on the maximization of social welfare:

$$\begin{aligned} &\underset{x_a \in X_a, x_0 \in X_0}{\text{minimize}} \quad f(x_0) \\ &\text{subject to} \sum_{a \in A \setminus 0} x^{\text{net}}_{a,t} = x_{0,t} \quad \forall\, t \in T \end{aligned} \tag{3}$$

where $x^{\text{net}}_{a,t}$ represents the net demand exchanged with the grid. The System-focused VPP, or VPP 1.0, focuses on maximizing operational efficiency and providing ancillary services, such as frequency regulation and voltage control.

Practical Example:

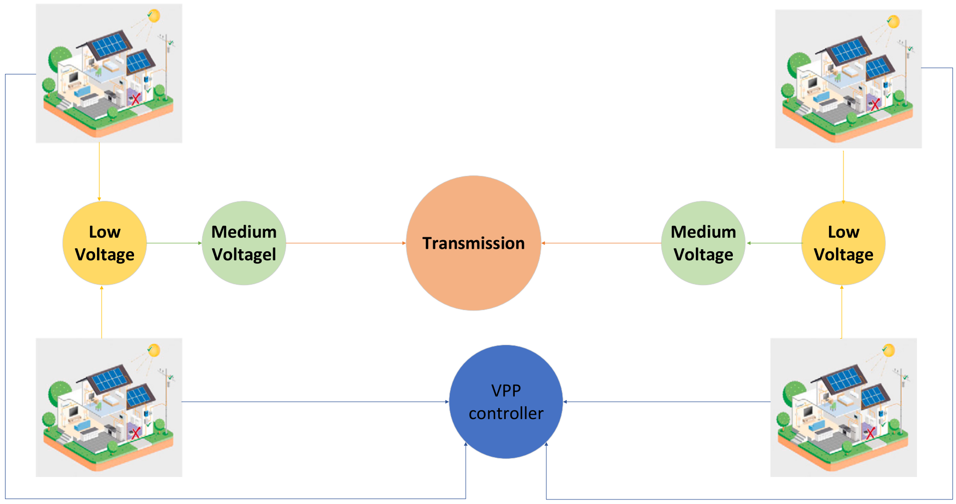

Fig. 3. Different Medium Voltage parts of the grid controlled by the VPP 0.0.

Consider a VPP 1.0 managing 30 wind turbines, each with a capacity of 2 MW, and a 10 MW solar plant. The goal is to provide ancillary services and participate in the wholesale market. Assume the maximum energy demand is 50 MW and the VPP can supply up to 80% of this demand.

The cost function for this scenario may include terms for frequency deviation penalties and transmission losses:

$$C(x_{0,t}) = 0.05 \times (x_{0,t})^2 + 0.02 \times x_{0,t} + 0.01$$

where $x_{0,t}$ is the total demand met by the VPP at a given time.

For a specific instance where the VPP meets 40 MW of demand, substituting the values into the cost function:

$$C(40) = 0.05 \times (40)^2 + 0.02 \times 40 + 0.01$$

$$C(40) = 0.05 \times 1600 + 0.02 \times 40 + 0.01$$

$$C(40) = 80 + 0.80 + 0.01 = 80.81 \text{ €}$$

Practical Example without VPP:

Consider the same scenario without the VPP, where individual producers directly interact with the grid without centralized control. Assume the cost of energy purchased from the grid is higher due to lack of negotiation power and absence of ancillary services, leading to additional penalties for frequency deviation and inefficiencies.

The cost function without VPP might be:

$$C_{no\_VPP}(x_0, t) = 0.07 \times (x_0, t)^2 + 0.03 \times x_0, t + 0.02$$

For meeting the same 40 MW demand:

$$C_{no\_VPP}(40) = 0.07 \times (40)^2 + 0.03 \times 40 + 0.02$$

$$C_{no\_VPP}(40) = 0.07 \times 1600 + 0.03 \times 40 + 0.02$$

$$C_{no\_VPP}(40) = 112 + 1.20 + 0.02 = 113.22\text{€}$$

This example highlights a cost reduction, showcasing the economic benefits of the VPP model. It is important to say that The VPP 1.0 model not only demonstrates higher efficiency in energy resource allocation, reducing unnecessary costs and improving system stability, but also provides ancillary services like frequency regulation and voltage control, which add value by preventing penalties and maintaining system reliability. By centralizing and optimizing energy generation and consumption, VPP 1.0 reduces operational costs, improves efficiency, and results in financial savings for its members. Additionally, participation in the wholesale market allows the VPP to negotiate better tariffs and obtain additional revenue, further benefiting its members.

2.2.3. Consumer-focused VPP or VPP 2.0

In the consumer-focused VPP or VPP 2.0 the main objective is to minimize the cost of the system and the consumer, and, because of that, it fits within the CVPP. Usually, it will present decentralized solutions, especially when the number of consumers reaches large scales because it allows greater flexibility and scalability. In a decentralized model, the decision-making is distributed across many DER owners, who can each make choices that optimize their interests. This reduces the burden on any central authority to coordinate all DERs and allows the system to be more resilient to failures and more adaptable to changes in demand. Considering this, to coordinate a great number of DERs, it can be used distributed and dual decomposition algorithms.

Distributed optimization algorithms are a type of algorithm designed to solve a global optimization problem by breaking it down into smaller sub-problems that can be solved independently by each participant. In a VPP, this means that each DER owner would solve a smaller sub-problem on their own, such as determining how much energy to generate or how much to charge for that energy. Once they have found a solution, the participants then communicate with each other to find a common solution that minimizes the overall cost of the VPP. This communication can happen in various ways, such as through a central coordinator or a peer-to-peer network. On the other hand, dual decomposition algorithms are another approach to decentralized optimization, where the global optimization problem is decomposed into smaller sub-problems, and each participant is assigned a sub-problem to solve. However, instead of communicating directly with each other, participants communicate with a central coordinator who aggregates the results and determines the optimal solution for the VPP.

This review will begin by discussing a centralized solution that is well-suited for VPPs with fewer participants. In such cases, a central authority can more easily coordinate and optimize the system due to the lower complexity and communication requirements.

- centralized solution

Whereas the user's electricity cost is given by:

$$g_a(x_a) = \sum_{t \in T} \sigma_t^{ft/tou} x_{a,t}^+ - \sigma_t^{fit} x_{a,t}^- \tag{4}$$

where $\sigma_t^{ft/tou}$ and σ_t^{fit} are the flat/ToU and the Feed in tariff respectively. $x_{a,t}^+$ and $x_{a,t}^-$ are, respectively, the energy imported from and exported to the grid.

Therefore, it can be described by

$$\underset{x a\in\mathcal{X}a, x0\in\mathcal{X}0}{\text{minimize}}\ f(x_0)+\gamma\sum_{a\in A\backslash 0} g_a(x_a)\ \text{subject to}\ \sum_{a\in A\backslash 0} x_{a,t}^{\text{net}} = x_{0,t}\ ,\ \forall\ t\in T \tag{5}$$

where the objective function consists of two terms: $f(x_0)$ and $\gamma\sum_{a\in A\backslash 0} g_a(x_a)$. The first term is a function of the decision variable x_0, while the second term is a sum of functions $g_a(x_a)$ of the decision variables x_a for each $a\in A\backslash 0$. In addition, the parameter γ represents a weighting factor that determines the relative importance of the two terms in the objective function. The problem is subject to a single constraint that specifies that the sum of the decision variables $x_{a,t}^{net}$ for all $a\in A\backslash 0$ at each time period $t\in T$ must be equal to the decision variable $x_{0,t}$.

- decentralized solution

The following method is more indicated when there are a lot of prosumers participating of the VPP [87]

$$\underset{x a\in\mathcal{X}a, x0\in\mathcal{X}0}{\text{minimize}}\ f(x_0)+\gamma\sum_{a\in A\backslash 0} g_a(x_a)+\sum_{t\in T}\lambda_t\left(\sum_{a\in A\backslash 0} x_{a,t}^{\text{net}}-x_{0,t}\right) \tag{6}$$

$$\text{Lagrange dual function:}\quad \underset{x_a\in\mathcal{X}_a, x_0\in\mathcal{X}0}{\text{minimize}}\ L_{\text{VPP}}(x,\lambda) \tag{7}$$

where $g_a(.)$ is the function of user, which could be adapted to integrate some properties in the model. It is possible to separate Lagrange dual function in

$$D(\lambda) := \underset{x_a\in\mathcal{X}a, x0\in\mathcal{X}0}{\text{minimize}}\left(D_0(\lambda)+\sum_{a\in A\backslash 0} D_a(\lambda)\right) \tag{8}$$

Solving by Distributed optimization using price coordination:

$$D(\lambda) := \underset{x_a\in\mathcal{X}_a, x_0\in\mathcal{X}0}{\text{minimize}}\ L_{VPP}(x,\lambda) \tag{9}$$

Considering that λ^k is the dual variable associated with the power balance constraint and represents the price update. The user agent solve can be represented by

$$D_a(\lambda) := \min_{\mathbf{x}a\in\mathcal{X}a} g_a(x_a)+\sum_{t\in\mathcal{T}}\lambda^k x_{a,t}^{\text{net}} \tag{10}$$

the VPP by

$$D_0(\lambda) := \underset{x0\in\mathcal{X}0}{\text{minimize}}\ f(x_0)-\sum_{t\in T}\lambda^k x_{0,t} \tag{11}$$

and the Lagrangian multiplier is updated by

$$\lambda^{k+1} = \lambda^k+\alpha^k\left(\sum_{a\in A\backslash 0} xa^{\text{net}}-x_0\right) \tag{12}$$

After some updates of the Lagrangian multiplier and interactions with the equations, it is possible to find a solution for the problem. Fig. 4 presents the function of this decentralized solution.

Practical Example:

Consider a VPP 2.0 with three DERs and three consumers, each generating energy through solar panels and consuming from the grid. Suppose the feed-in and consumption tariffs are 0.05 €/kWh and 0.15 €/kWh, respectively. If consumer A generates 10 kWh and consumes 8 kWh in a given period, their cost function $g_A(x_A)$ would be calculated as:

$$g_A(x_A) = (8\ \text{kWh}\times 0.15\ €/\text{kWh})-(2\ \text{kWh}\times 0.05\ €/\text{kWh})$$

$$g_A(x_A) = 1.20\ €-0.10\ € = 1.10\ €$$

For the decentralized optimization, assume λ is the price update variable:

$$D_a(\lambda) := \min_{\mathbf{x}_a\in\mathcal{X}_a} g_a(x_a)+\sum_{t\in\mathcal{T}}\lambda x_{a,t}^{\text{net}}$$

$$D_0(\lambda) := \underset{x_0\in\mathcal{X}_0}{\text{minimize}}\ f(x_0)-\sum_{t\in T}\lambda x_{0,t}$$

The Lagrangian multiplier is updated iteratively to find the optimal cost.

Now, let us consider the overall VPP cost. Suppose we have three consumers A, B, and C. Their individual costs can be summed up:

$$\text{Total VPP Cost} = g_A(x_A)+g_B(x_B)+g_C(x_C)$$

Assume the following for simplicity: - Consumer B generates 15 kWh and consumes 10 kWh:

$$\begin{aligned} g_B(x_B) &= (10\ \text{kWh}\times 0.15\ €/\text{kWh})-(5\ \text{kWh}\times 0.05\ €/\text{kWh}) \\ &= 1.50\ €-0.25\ € = 1.25\ € \end{aligned}$$

- Consumer C generates 20 kWh and consumes 12 kWh:

$$\begin{aligned} g_C(x_C) &= (12\ \text{kWh}\times 0.15\ €/\text{kWh})-(8\ \text{kWh}\times 0.05\ €/\text{kWh}) \\ &= 1.80\ €-0.40\ € = 1.40\ € \end{aligned}$$

The total cost for the VPP 2.0 would be:

$$\text{Total VPP Cost} = 1.10\ €+1.25\ €+1.40\ € = 3.75\ €$$

Practical Example Without VPP:

Consider the same scenario without the VPP, where individual producers directly interact with the grid without centralized control. Assume the cost of energy management is higher due to lack of optimization and higher penalties for not adhering to optimal power flow. The cost function without VPP might be:

For consumer A:

$$\begin{aligned} g_A(x_A) &= (8\ \text{kWh}\times 0.18\ €/\text{kWh})-(2\ \text{kWh}\times 0.03\ €/\text{kWh}) \\ &= 1.44\ €-0.06\ € = 1.38\ € \end{aligned}$$

For consumer B:

$$\begin{aligned} g_B(x_B) &= (10\ \text{kWh}\times 0.18\ €/\text{kWh})-(5\ \text{kWh}\times 0.03\ €/\text{kWh}) \\ &= 1.80\ €-0.15\ € = 1.65\ € \end{aligned}$$

For consumer C:

$$\begin{aligned} g_C(x_C) &= (12\ \text{kWh}\times 0.18\ €/\text{kWh})-(8\ \text{kWh}\times 0.03\ €/\text{kWh}) \\ &= 2.16\ €-0.24\ € = 1.92\ € \end{aligned}$$

The total cost without VPP would be:

$$\text{Total Cost Without VPP} = 1.38\ €+1.65\ €+1.92\ € = 4.95\ €$$

In conclusion, the VPP 2.0 model shows significant cost savings:

$$\text{Savings} = 4.95\ €-3.75\ € = 1.20\ €$$

2.2.4. VPP 3.0: Optimal power flow problem

It is also possible to transform the two previous VPPs (1.0 and 2.0), which aim to solve optimization problems, into VPPs linked to the optimal power flow problem. This new category is called VPP 3.0 [7,83–86]. It takes into account not only the network constraints, being networked aware, but also the customers, since with the optimal power flow solution there is a large cost reduction. For those reasons, this category is at the intersection of CVPP and TVPP. It can be mathematically described as follows:

$$X = X_a\cup X_n,\ x_0\quad\in X_n\ \underset{xn\in\mathcal{X}n, xa\in\mathcal{X}a}{\text{minimize}}\ f(x_0)+\gamma\sum_{a\in A\backslash 0} g_a(x_a) \tag{13}$$

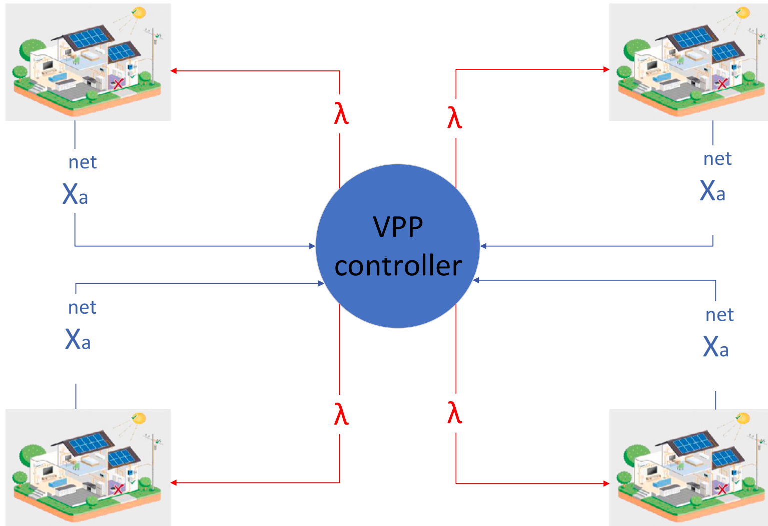

Fig. 4. VPP 2.0 decentralized solution scheme.

the power balance constraints $\forall i \in N, \quad p_{g,i} = q_{g,i} = 0 \quad \forall i \in N \setminus 0$ are:

$$p^g_{i,t} - p^d_{i,t} = v_{i,t} \sum_{j \in N} v_{j,t}(g_{ij} \cos \theta_{ij,t} + b_{ij} \sin \theta_{ij,t}) \tag{14}$$

$$q^g_{i,t} - q^d_{i,t} = v_{i,t} \sum_{j \in N} v_{j,t}(g_{ij} \cos \theta_{ij,t} + b_{ij} \sin \theta_{ij,t}) \tag{15}$$

voltage and power constraints:

$$\underline{v} \le v_{i,t} \le \overline{v} \tag{16}$$

$$\underline{p} \le p_{i,t} \le \overline{p} \tag{17}$$

$$\underline{q} \le q_{i,t} \le \overline{q} \tag{18}$$

where $\underline{v}/\overline{v}$ represents the maximum/minimum voltage at a bus, $\underline{p}/\overline{p}$ represents the maximum/minimum active power injection at a bus, and $\underline{q}/\overline{q}$ represents the maximum/minimum reactive power injection at a bus. The problem is not decomposable, so two copies of powers injected will be created:

$$\hat{p}_{i,t} = p_{i,t}, \ \{\forall \ a \ \epsilon \ A \backslash 0 | \ A \subseteq N\}, \ \forall \ t \ \epsilon \ T$$

$$\underset{xn \in \mathcal{X}n, xa \in \mathcal{X}a}{\text{minimize}} \quad f(x_0) + \gamma \sum_{a \epsilon A \backslash 0} g_a(x_a) \quad \text{subject to} \quad p_a = \hat{p}_a \ , \ \forall \ a \ \epsilon \ A \tag{19}$$

where $\hat{p}_i$ is the duplicate variable of p_i at bus i. In view of all of this, it is possible to obtain the solution by Alternating Direction of Multipliers Method(ADMM) [88], which will decompose the large-scale problem into local sub-problems and solve them in parallel for the agents. The augmented Lagrangian can be described as follows

$$L(x_n, x_a, \lambda_a) := f(x_n) + \sum_{a \in A \backslash \{0\}} g_a(x_a) + \sum_{t \in T} (\lambda_{a,t}(p_{a,t} - \hat{p}_{a,t}) + \frac{p}{2}(p_{a,t} - \hat{p}_{a,t})^2) \tag{20}$$

in it, p is a penalty parameter and $\lambda_a := \left\{ \{\lambda_a\}_{a \in A \backslash 0} \right\}_{t \in T}$ represent the dual variables with the coupling constraints as explicit in the previously mentioned articles. After that, it is possible to decompose the problem in into the ADMM general form

$$x^{k+1}_n := \underset{\mathbf{x}_n \in \mathcal{X}_n}{\arg\min} \ L(x_n, x^k_a, \lambda^k_a) \tag{21a}$$

$$x^{k+1}_a := \underset{\mathbf{x}_a \in \mathcal{X}_a}{\arg\min} \ L(x^{k+1}_n, x_a, \lambda^k_a) \quad \forall \, a \in A \setminus \{0\} \tag{21b}$$

$$\lambda^{k+1}_a := \lambda^k_a + \rho \left(p^{k+1}_a - \hat{p}^{k+1}_a \right) \quad \forall \, a \in A \setminus \{0\} \tag{21c}$$

where the last equation updates the value of x_n for the next iteration, by finding the argument that minimizes the Lagrangian function L with respect to x_n. The second updates the value of x_a for the next iteration, using the new value of x_n computed in the previous step, by minimizing L with respect to x_a for each a in the set A except 0 and the third updates the Lagrange multipliers $lambda_a$ for the next iteration, using the current values of p_a and $\hat{p}_a$, which are related to the primal and dual residuals of the optimization problem.

Practical Example:

Suppose a VPP 3.0 integrates 50 MW of solar energy, 30 MW of wind energy, and 20 MW of battery storage. The network has line capacity constraints of 100 MW and voltage constraints of 220 V to 240 V.

The power flow optimization cost function can be expressed as:

$$\min \sum_{t \in T} \left[a \cdot (p_t)^2 + b \cdot p_t + c \right]$$

with power constraints:

$$\underline{p} \le p_{i,t} \le \overline{p}$$

and voltage constraints:

$$\underline{v} \le v_{i,t} \le \overline{v}$$

If, in each period, the VPP supplies 80 MW of energy with a 50% solar and 50% wind composition, the cost function would be:

$$\text{Cost} = 0.01 \times (80)^2 + 0.05 \times 80 + 1$$

$$\text{Cost} = 64 + 4 + 1 = 69 \text{ €}$$

Practical Example Without VPP:

Consider the same scenario without the VPP, where individual producers directly interact with the grid without centralized control. Assume the cost of energy management is higher due to lack of optimization and higher penalties for not adhering to optimal power flow.

The cost function without VPP might be:

$$\min \sum_{t \in T} \left[0.02 \cdot (p_t)^2 + 0.07 \cdot p_t + 2 \right]$$

For meeting the same 80 MW demand:

$$\text{Cost} = 0.02 \times (80)^2 + 0.07 \times 80 + 2$$

$$\text{Cost} = 128 + 5.6 + 2 = 135.6\ €$$

The advantages of VPP 3.0 are numerous and significant. Implementing VPP 3.0 demonstrates substantial cost savings and highlighting the economic benefits of the VPP model. VPP 3.0 optimizes power flow, ensuring efficient energy distribution and minimizing losses, which contributes to overall system efficiency. Additionally, the VPP 3.0 model effectively integrates network constraints and customer needs, providing a balanced solution that benefits both the grid and consumers. The ability to provide ancillary services such as voltage control and frequency regulation adds significant value by maintaining grid stability and reliability.

2.3. Other Virtual Power Plant categories

Although the concepts previously defined are comprehensive with all categories, other definitions exist as a way of grouping even more similar VPPs. In this subsection, some of these different classifications will be presented.

2.3.1. Electrical vehicle-Virtual Power Plant

The first modality is a VPP that mostly or only aggregates electric vehicles, being called EV-VPP. And within this set, depending on the nature of the participants, it can be divided into two subsets: plug-in EV-VPPs, which includes pure EVs with PHEVs, and fuel cell EV-VPPs [89,90].

It represents an interesting solution not only because a disorderly connection of several EVs in the network would easily cause problems, such as those related to peak–valley or supply–demand [91], but also because an organization of these technologies can be highly beneficial to the owners and the network, especially in providing the most varied types of services to the grid [92,93]. It stands out for regulating voltage in the utility grid, managing frequency deviations [94], adjusting voltage fluctuations, and balancing supply–demand. They also store energy, helping the users to participate in the market and arbitrate their expenditures most conveniently.

There are several ways to carry out the Vehicle- grid integration (VGI), highlighting the following: Vehicle-to-building concepts (V2B), Building-to-vehicle-to-building (V2B2), Vehicle-to-grid (V2G), Vehicle-to-home (V2H), Vehicle-to-load (V2L), Vehicle-to-everything (V2X), Vehicle-to-vehicle (V2V), Vehicle for grid (V4G).

2.3.2. Dynamic Virtual Power Plant

The second category presented has great relevance, due to the worldwide trend of increasing the use of RES. It is called Dynamic Virtual Power Plant (DVPP) consists of a VPP formed only by clean sources chosen and organized in such a way that they manage to deal with the intermittence very well, having an increase in the reliability of the whole as a big consequence [95]. Because of that, they can participate in the wholesale market and provide ancillary services to the grid with consistency and strength [96]. The greater the geographic distribution, the better the results of this class of VPP must be since the individual systems can suffer more sudden and unpredictable variations, however, the set can continue to provide what was expected due to the variety of climatic conditions [97].

They are generally composed of solar photovoltaic power plants, solar thermal power plants, including thermal energy storage in molten salts, offshore or onshore wind power plants, hydropower power plants, pumped-storage hydropower with the bidirectional operation, biomass power plants, geothermal power plants and additional units like batteries, hydrogen electrolyze and others.

2.3.3. Fog as a Virtual Power Plant

The last category presented by this paper is the use of the fog system as a VPP, and this association is called Fog as a Virtual Power Plant (FaaVPP). Fog is a cloud-like system, in other words, it is a system that stores and manages a large amount of complex data. However, the fog has a difference compared to the cloud: the data is stored on a system close to where the device came from, being more secure.

This model is more centralized and, although it can use centralized schemes, it normally involves distributed control, as it uses a network of edge devices (fog nodes) to manage energy resources and coordinate their activities, each fog node being able to operate autonomously. For this reason, many authors believe it to be more viable and organized than decentralized systems. In FaaVPP, programming, such as Linear Programming (LP), can be used to associate with rules that stimulate the community when a beneficial behavior happens with frequency. For instance, in [98], when more prosumers interacted with one another they could get better rates of energy. Although this type of aggregator requires a lot of advanced technologies and a well-developed organization, it presents itself as a good solution to manage energy and provide ancillary services too [99].

3. Technology of virtual power plants

Different state-of-the-art technologies enable the VPP, including various distributed generation units, energy storage systems, communication technologies, etc.

3.1. Distributed energy resources

Distributed energy resources (DERs) refer to a variety of small-scale energy resources that can be connected to the local power grid or operate independently in isolated applications. This concept has gained popularity in recent years, due to the cheaper technologies involved, technological advances, benefits to the environment, and energy security. This review will consider DERs such as distributed generation (DG), energy storage systems (ESSs), electric vehicles (EVs), smart loads, small hydropower plants, and other renewable energy sources.

These components play an important role in providing electricity to distributed grids and can increase distribution efficiency by integrating more renewable energy sources into the system [100]. They can also contribute to network rebuilds to help stabilize the distributed system, reducing voltage dips and sags [101]. In addition, DERs can provide a number of benefits to the grid, such as improving resiliency by providing power during power outages, meeting peak demand, providing ancillary services such as frequency reserve and reactive power, and reducing operating costs.

3.2. Renewable energy sources

RES are sources that sustainably produce energy from natural conditions and do not emit any amount of greenhouse gases during energy generation, some examples are solar, wind, hydro, and ocean energy. They have been gaining more relevance due to the greater need for energy in the current decade, limits on fossil fuel reserves, concern for environmental issues, technological advances, etc.

Recent years have witnessed notable experimental progression in these systems. Solar panels, for instance, experienced notable improvements in their efficiency and a reduction in production costs. Additionally, advances in perovskite solar cells, bifacial solar panels, and floating solar farms have also contributed significantly to the evolution of RES [102]. Wind power has also seen major changes: turbines have become more efficient, with some models now capable of producing over 10 MW of power. The growth of offshore wind farms is noteworthy due to their ability to capture stronger and more consistent winds compared to onshore wind farms [103]. Floating wind turbines are also being developed and offer the potential to expand wind energy production to deeper waters. Hydropower and ocean power have also brought significant changes, such as new designs of improved turbines for hydropower, development of Pumped storage hydropower (PSH) storage systems, the use of hydrokinetic energy to generate power from ocean currents, tide and river currents [104]. It is worth mentioning the advances in tidal energy turbine conversion technologies and ocean thermal energy conversion technology [105].

The increasing integration of renewable energy sources, especially wind farms and photovoltaic systems, plays an important role in the VPPs due to the capability of these groups to optimize generators and minimize damage caused the stochastic and intermittent characteristics of the RESs. These stochastic generation units are also non-dispatchable, however, they should be backed up with Energy Storage Systems (ESSs), mainly to mitigate their intermittency and ensure a stable and reliable energy supply [49].

3.3. Energy storage systems

ESSs play a crucial role in balancing energy production and supply. The distributed generation units are mostly renewable energy sources, in which energy storage systems contribute to attenuate the frequency fluctuation caused by the intermittent generation of various RESs connected to the grid. Furthermore, ESSs are widely applied for power grid frequency adjustment, power flow optimization, etc. [106].

ESSs are also important in implementing clean sources and improving grid stability. The research and development of technologies for energy storage systems is a broad and dynamic field, ranging from pumped storage hydropower, thermal, pressure storage, chemical energy storage, electrochemical, etc. The latest innovations and future trends in this area can be explored in depth in the following work [107]. However, the present review will consider batteries as the most commonly used energy storage systems as they are the most common ESSs in the everyday lives of prosumers, and because the continuous research and development of new materials such as electrolytes solid-state batteries and new chemistries such as lithium-sulfur and lithium-air offer prospects for more efficient, durable and cost-effective battery solutions.

ESSs play a vital role in dealing with intermittency by storing excess energy generated during periods of low demand and releasing it during periods of high demand. As the deployment of RES continues to increase in VPPs, the integration of energy storage systems will become even more crucial to ensuring grid stability and reliability. In addition, the integration of advanced technologies such as artificial intelligence and machine learning algorithms is also important. These technologies can optimize the performance of ESSs by predicting energy demand and adjusting battery charge and discharge cycles, resulting in greater efficiency and longer battery life.

The ESSs used in the VPPs are categorized into different types based on the utilized technologies, shown in Fig. 5. In addition, the discharge time and the rated capacity for different energy storage technologies can be referred to [108].

Moreover, electrical vehicles (EVs) are also considered energy storage units in VPPs [89]. In [109], a fleet of EVs is utilized as an energy reserve to coordinate with the wind farm in VPPs to optimize the operation cost. The coordination of EVs with battery-ultracapacitor (UC) to mitigate the fluctuations of wind power for a VPP is presented [110]. The EV is utilized to improve the operation efficiency of VPP and the influence of EVs on the bidding strategies of VPP is investigated considering different types and scales in [111]. In addition, the EVs applied to balance the energy for the power grid and VPPs, are reviewed in [90], where different integration modes of EVs in the VPPs are also discussed including employing them as energy buffers similar to energy storage systems. In particular, the EVs are combined with different DERs in the VPPs to reduce the impact of EV large-scale grid connections, and fluctuations from the stochastic RES, providing ancillary services, etc. to realize economic and sustainable operations for the VPP.

3.4. Demand response/controllable loads

Demand Response (DR) is a term used for Demand Side Management (DSM) programs designed to encourage end-users to make short-term changes in energy demand in response to a price signal from the electricity hourly market, or a trigger initiated by the electricity grid operator. The different DR techniques utilized for reshaping load profiles mainly consist of peak clipping, valley filling, load shifting, strategic conservation and load growth, and flexible load shape [112]. And the controllable loads contribute to the flexibility and controllability of the system, which are categorized into three groups: residential, commercial, or municipal consumers [62].

With the VPPs' control, DR-based decisions can adjust the energy consumption patterns and loads to manage the energy supply and demand inside the VPP based on real-time energy prices. The controllable loads also facilitate the management of VPP according to the changing energy consumption or energy prices. In this way, the VPP can also utilize the DR techniques combined with RESs as an effective reserve service to neutralize its volatility.

3.5. Information and communication technology

Virtual Power Plants (VPPs) present the excellence of Information and Communication Technology (ICT) in the energy sector. They serve as a versatile hub that orchestrates energy generation, distribution, transmission, consumption, storage and retail operations.

At the core of VPP's functionality is a sophisticated control and management system, facilitated by cutting-edge communications technologies [113]. These technologies enable smooth coordination between multiple DERs, ESSs, loads, and other components. This coordination can occur through one-way and two-way communication channels, ensuring an optimal flow of information [21].

The operational complexities of the entire system, including the real-time status of each sector, condition monitoring, in-depth analysis, and accurate forecasts, are efficiently transmitted through the ICT infrastructure. Furthermore, this data is securely stored, whether in the cloud or on edge computing devices, ensuring accessibility and reliability for critical decision-making processes. VPPs are one of the typical demonstrations of Information and communication technology (ICT)-embedded systems in the energy sector.

The coordination among different DERs, ESSs, loads, etc. is realized via the control and management system relying on the communication technologies in either a unidirectional or bidirectional way. The information of the system operation including the status of each sector, condition monitoring, analysis and forecast, optimization command, etc. is transmitted with the ICT and stored in the cloud or edge computing devices [114].

3.6. Advanced technologies

3.6.1. Intelligent control

With the development of artificial intelligence, it is widely applied in different areas including the smart energy system [115]. It can coordinate different components inside the system and also facilitate the management among different participants of the system based on the data collected from different fields to provide optimal operation, maintenance, control, etc. With AI analyzing the power generation, supply, and consumption of VPPs, it eases energy management using deep learning, reinforcement learning, and the like machine learning technologies based on the electricity market, demand, and historical operational data. The applications of intelligent control in the VPPs are summarized below

- Reinforcement learning (RL). The reinforcement learning learns optimal policy by interacting with the system [116]. In particular, the RL agent observes the states of the system such as energy

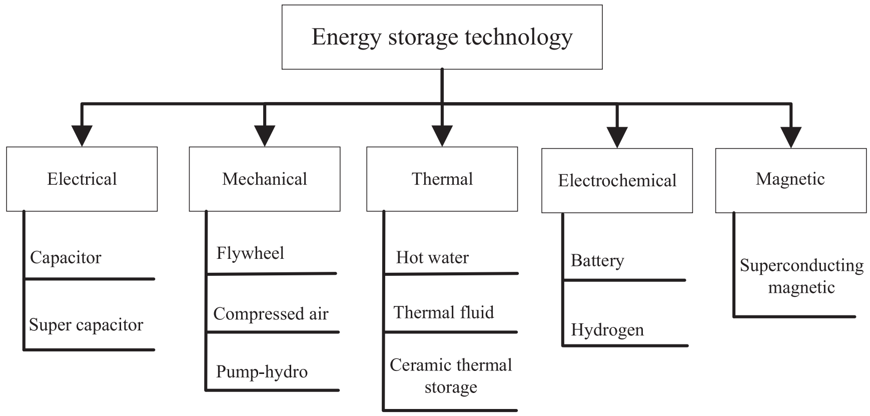

Fig. 5. Different types of energy storage systems.

prices, battery state-of-charge (SoC), load demand, etc., to take optimal actions for different defined optimization objectives, such as minimization of energy cost, battery degradation, etc. The actions could be bidding strategy on an energy market, dispatch strategies for batteries, etc., which changes the operation of the system [117]. A reward would be returned to the RL agent, who learns to take action to obtain more rewards in the long run during the training process.

- Regression method. The regression method is one of the Machine Learning (ML) characteristics for mapping the relationship between the input–output pairs. In VPP, time-series forecasting, such as load forecasting, load prediction, etc., is a typical application scenario. Based on the training dataset, the method is usually employed for predictions of variables inside the system. In addition, it is also employed for scheduling problems to obtain optimal operations [118].
- Classification. Another important application of ML is classification. It is usually applied to detect or diagnose a fault state of the sectors in the system. For supervised learning, based on the labeled fault dataset, the fault types could be identified. While unsupervised learning is usually used for fault diagnosis [119].

3.6.2. Edge computing

In the future virtual power plants, numerous data are collected through the smart devices in real-time, which are transmitted to the central controller for optimizing the dispatch strategies of different DERs. Due to the centralized control, the large amount of data would introduce a burden to the communication infrastructure, especially for transmission interference, limited storage capacity, etc. In addition, the uncertainty caused by the stochastic large-scale solar and wind power brings challenges of scalability and adaptability of centralized controllers. Therefore, decentralized control is explored for providing flexibility in VPPs [120].

To address these, edge computing is employed to ease the communication burden and also improve the scalability [117,121]. The massive data generated and collected in the DERs are handled in the edge nodes in the cloud environment respectively in different sectors of VPP. In addition, it can facilitate and accelerate data processing to achieve real-time economic dispatches. The storage and computation, as well as the privacy of end users, can be addressed with edge computing.

3.6.3. Smart inverters

The smart inverters, interfacing between energy generation and consumption, offer different features for system operations. The major features of smart inverters are summarized below:

- Cooperativeness. The inverter could function according to the operations of other components of the VPP, such as cooperation of multiple inverters of fast dynamics such as oscillation suppression [122], vol/var control (VCC) [123].
- Autonomy. Due to the geographically dispersed DERs in VPP, though the communications among them enable the proper coordinated controls, to reduce the cost for each control aspect, the inverter should maintain autonomous characteristics to be able to function properly under specific conditions, such as power flow control [124], inertia support [125], etc.
- Self-adaptability. Another important characteristic of the smart inverter is the adaptability to varying operating conditions, which involves grid impedance estimation, frequency self-synchronization, etc. In [126], an artificial neural network is employed to estimate the grid impedance so that the system stability is guaranteed. In addition, intelligent fault-tolerant control ensures the reliable operation of the system.
- Self-awareness. The self-awareness feature of smart inverters could be categorized into prognosis, condition monitoring, and diagnostics. The diagnosis involves fault detection, which is usually caused by the thermomechanical fatigue of IGBTs inside the converter [127], such as open-circuit faults, short-circuit faults, etc. The condition monitoring is to evaluate the component health status in real-time to ensure the reliable and stable operation of the inverter, including capacitor, switching devices, heat sinks, etc. [128]. The prognosis is to predict the potential faults, facilitating the maintenance of the inverter, which is usually realized along with the condition monitoring [129].
- Plug and play. The inverters are required to be compatible with the standard communication protocols so that control the VPP. Different types of control for the VPP and the corresponding communication requirements are presented in the next sections. Therefore, to enable the smart inverter for the VPP, communication technologies should be developed properly. Different communication technologies for smart grids are reviewed in [130].

3.6.4. Blockchain and smart contracts applied to peer-to-peer energy trading

A Peer-to-Peer (P2P) Transactive Energy Market (TEM) can be considered a decentralized energy market where participants trade electricity directly with each other without the need for an intermediary [8]. This allows people to play active and important roles in the power grid and gives them greater freedom to decide how to use their energy production. Usually, P2P uses digital platforms that connects buyers and sellers and enables them to trade energy in real time. The platform typically uses blockchain technology, smart contracts, and advanced metering systems to ensure transparency, security, and accuracy of energy transactions [65,131].

The blockchain is built on the linking of different blocks, which consists of a block header, transaction information, and a hash. The three types of nodes in a blockchain, mining nodes, super nodes, and light nodes, are fundamentals to the network, where the nodes back up all or part of the data from a distributed ledger. The data can be reconstructed based on other nodes if one of the nodes is attacked, which ensures security. Due to the characteristics of blockchain, the secure and decentralized management of P2P transactions among multiple units in the VPPs can be realized [61,132].

Although P2P energy trading can present several challenges and problems as seen in Section 1.1.4, it can offer advantages and incentives for VPPs over traditional markets [133]. First, it increases the penetration of renewable energy sources in the energy producers by encouraging the deployment of DERs. Second, P2P trading provides consumers with greater freedom over their energy by allowing them to choose their energy sources and negotiate prices directly with other participants. Third, P2P trading can reduce the reliance on centralized energy systems and improve the resilience and reliability of the grid. Fourth, encourages prosumers to participate in VPPs seeking the strengthen their production. Fifth, it represents a mean of organizing the sets in a way that does not remove the freedom of the participants as occurs in centralizing solutions.

4. Organizational control of virtual power plants

Conventional power plants participate in various markets by supplying energy and providing ancillary services, such as day-ahead market [134], real-time balancing market [135], etc. In particular, the VPP manages the different DER units to participate in these markets, achieving optimal coordination with the system operators, especially with the increasing number of DER in the distribution network. To coordinate different energy sectors in the VPPs, a proper energy control strategy is critical for the economic operation of VPPs. The different control methods of VPPs could be categorized as follows.

Fig. 6 presents a detailed overview of four distinct control structures for VPPs, each tailored to address specific operational challenges and management strategies. In the centralized control structure (Fig. 6a), the grid operator exerts direct, overarching control over all DERs within the VPP. This model centralizes decision-making, allowing for a cohesive and unified strategy across the entire system. All relevant data, such as load demands, generation outputs, and market transactions, are funneled to a central control point, which then issues commands to each DER. While this ensures a high degree of coordination and alignment with market dynamics, it also introduces significant computational and communication burdens as the number of DERs increases. Additionally, this structure may pose challenges related to data privacy and the complexity of managing geographically dispersed resources.

In contrast, the decentralized control structure (Fig. 6b) decentralizes decision-making by distributing control responsibilities to local operators or coordinators within each sector of the VPP. These local entities manage their respective DERs and communicate with a central gateway that interfaces with the grid operator. The decentralization allows for more localized decision-making and reduces the computational load on a single central entity. However, it introduces potential inefficiencies due to the lack of a global perspective, which can lead to suboptimal coordination between sectors and challenges in achieving overall system optimization.

The distributed control structure (Fig. 6c) takes decentralization a step further by enabling direct communication and coordination between local operators without the need for a central gateway or control point. This peer-to-peer communication model enhances system resilience by eliminating single points of failure and allowing for more adaptive and scalable operations. Each sector or DER can independently adjust its operations based on real-time information from neighboring sectors. However, this structure's reliance on robust and secure communication networks makes it vulnerable to cyber-attacks and requires careful management to prevent information silos and ensure consistent operation across the entire VPP.

Lastly, the comprehensive control structure (Fig. 6d) integrates the strengths of both centralized and distributed control approaches. In this hybrid model, the VPP is managed centrally at a high level, but operational execution is delegated to local operators, allowing for both global coordination and local flexibility. This two-tiered approach mitigates the computational and storage burdens on the central controller while still maintaining a cohesive strategy across the VPP. However, the need for continuous and reliable communication between the central and local levels introduces potential vulnerabilities, particularly concerning cyber security and communication bandwidth constraints.

Each of these control structures offers distinct advantages and disadvantages, as summarized in the accompanying figure and table. The choice of control strategy must consider the specific needs of the VPP, including the scale of operations, geographic distribution of resources, and the importance of privacy and security in the control process.

4.1. Centralized control

In the centralized control structure, shown in Fig. 6(a), the components of the VPPs are controlled via the Control Coordination Center (CCC), which is also responsible for the transactions between the markets. The centralized control is usually applied to VPP 0.0, where central control have full control of the energy sectors of the system. In general, all the required signals such as load signals, energy generation, etc., are sent to the control center, and the corresponding command signals are sent to an independent controller for each DERs. In this way, the centralized control method can achieve ultimate power control of different units by coordinating the power generation and consumption [49]. Because of this, the centralized control mode keeps consistent with the market, which ensures the security of the system. In addition, it can effectively protest the interests of all parties in the system with unified standards, management, and operations. In [136], a centralized bidding control strategy is proposed to optimize the economic benefits of the VPP system by full control of the DERs. In [137], a centralized, coordinated frequency control strategy for DERs is proposed for the VPP, which is to ensure a consistent response from all the resources in the VPP. A centralized control structure is adopted in [121], which is responsible for system dispatching, including energy purchase and sales, dealing with RESs uncertainty, energy balance, obtaining information signals from energy conversion units to optimize their operations, etc.

However, with the increasing aggregated sectors in the system, the scheduling for all the units becomes complex, which in turn increases the computational burden. Moreover, the DERs in the VPPs are different in terms of property rights [138] and time-varying status [139], which causes great uncertainty to the operation of the system. Moreover, due to the geographically dispersed DERs, different subregions may not share the information with others for their interest, which would induce privacy issues. All of the above contributes to the high requirement for communication bandwidth and central storage for information transmission and gathering.

In short, for the centralized control structure, VPP is responsible for scheduling optimization for different units of the system, which requires sufficient communication bandwidth for information collection and command transmission. The computation burden with the increasing number of units also limits its application in VPPs control structure.

4.2. Decentralized control

In a decentralized control structure, shown in Fig. 6(b), a virtual coordinator is necessary to coordinate the operations of different units in the top layer due to the lack of direct communications between units. As mentioned, the decentralized control is usually employed for the

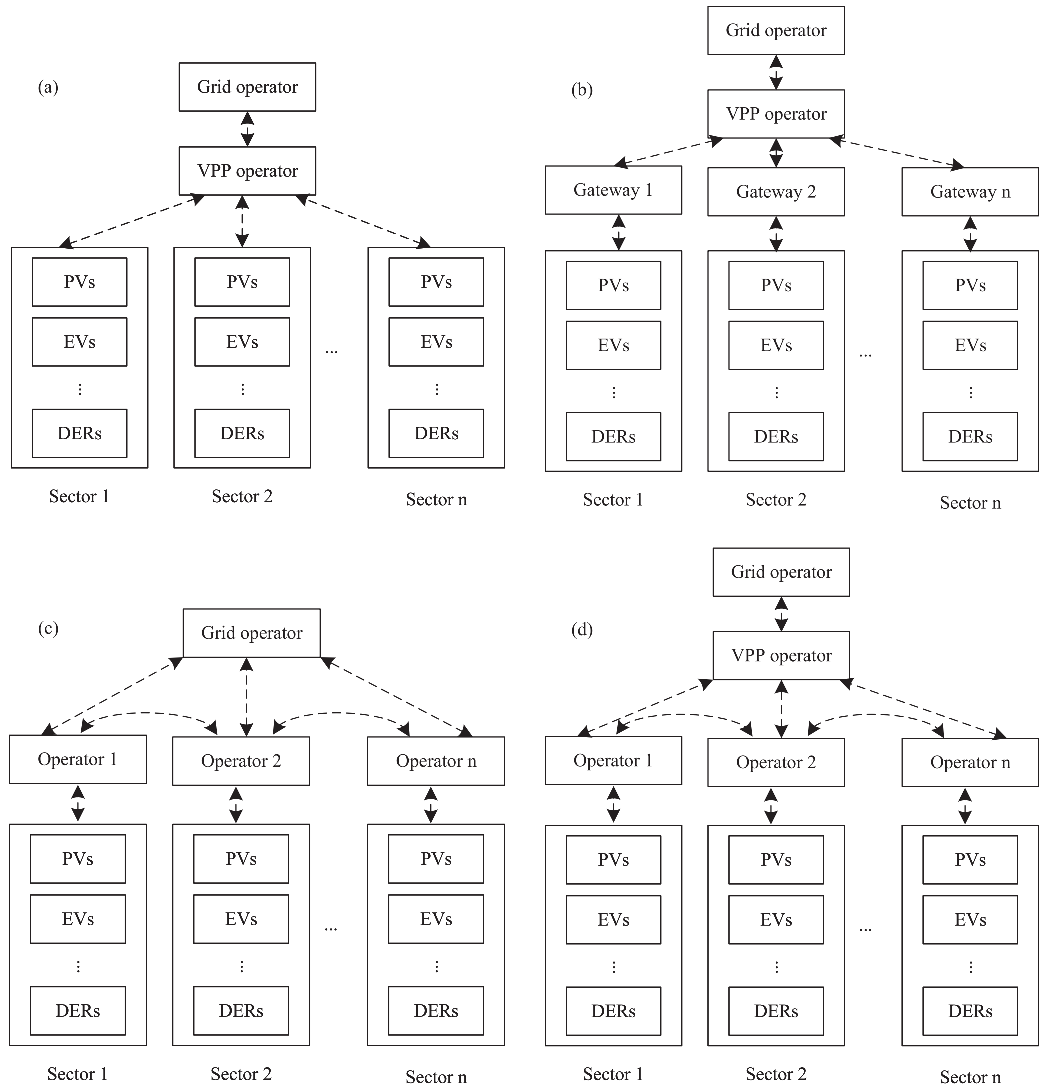

Fig. 6. Different control structures for VPP. (a) centralized control, (b) decentralized control, (c) distributed control, and (d) Comprehensive control.

control of VPP 2.0 and VPP 3.0, where the local controllers make decisions for the DERs and compete among each others for more profits. In addition, the DER units and the middle entities communicate with the center layer entities in star networks [140]. In addition, the decentralized control structure can distribute the computation burden, storage, and communication requirements, thus easing the requirement of the coordinator in the top layer compared with the central controller in the centralized controller. In particular, the decentralized methods consist of three types, iterative information exchanged-based method [120], game-theoretic method [141] and auction-based method [142].

The conventional decentralized methods require complete information and assume that the participants in the system make rational decisions. In addition, for the application of VPP, the strong coupling relationships among the coordinator and controlled objects leads to complexity and difficulty in the control of VPPs. To solve the problem, a bottom-up approach for the flexibility exploitation of VPPs is proposed in [120], which is built based on the autonomous decentralized system theory. In such a structure, the DERs are highly autonomous and interact while remaining independent of each other. Therefore, the independent DERs have no impact on the system operation. In this way, the DERs can be aggregated and expanded online in the VPPs. In [143], a blockchain-based decentralized energy management strategy is proposed for DERs in VPP. To improve the fault tolerance, adaptability, and privacy awareness of VPPs, in [144], a decentralized structure based on a message queue (MQ) is employed, which improves the calculation and communication efficiency while maintaining the privacy and fault-tolerant characteristics.

In addition, the decentralized control structure can distribute the computing and data storage burden to the local operators, thus the requirements for the top entities are relieved compared with the centralized controller in the CCC [145]. In [146], a decentralized prediction provision algorithm is proposed to overcome the challenges of centralized control including the extensive data processing and potential violation of information privacy. On top of these, the decentralized control algorithm aims at achieving economic operations for each consumer.

In a conclusion, decentralized control could distribute the computing burden and storage requirement to local controllers.

4.3. Distributed control

As the geographically dispersed DGs are aggregated into the VPPs, the distributed control strategy becomes popular and is utilized to coordinate different sectors in VPPs for achieving optimal operational cost, shown in Fig. 6(c). The control structure is mostly applied in FaaVPP and other larger scale VPP due to the involvement of numerous energy units. It can not only protect the privacy of each independent unit inside the system but also alleviate the computational burden and storage requirements for the control system. In particular, the distributed control structure is divided into two independent layers, of which the first is the central controller of VPP and the second is the controller operated by the local stakeholders. The neighboring local operators communicate with each other and exchange information to achieve economic optimization for each individual.

A coordination control strategy for short-term frequency containment to improve the dynamic response and maintain the frequency stability of the system is proposed [147]. In [148], a distributed real-time clustering algorithm is proposed to dynamically aggregate the energy storage systems into the VPPs. It also cooperates with distributed secondary controller for controlling the ESSs of each VPP so that a control structure that reduces the power losses and improved voltage regulation is built. Similar to the decentralized control method, the computational burden is also distributed among the local edge devices. In addition, the main advantages of the distributed control structure compared with the centralized control include plug-and-play scalability, elimination of single points of failure, etc.

However, due to the communication requirement of distributed control, the cyber–physical system is susceptible to cyber-attacks. To solve the issue, in [149], a secure distributed economic scheduling method is proposed by isolating the attacked units through monitoring among the neighboring units. In addition, the distributed control is based on communications between the neighboring units, which depends on the strongly connected cyber–physical graph. However, in a practical system, the communication connectivity needs to be verified to ensure the feasibility of the distributed control and avoid information islands, thus increasing the additional burden on the system.

4.4. Comprehensive control

The comprehensive control method is the combination of centralized control and distributed control methods, which incorporates the characteristic of the centralized and distributed controller, shown in Fig. 6(d). The control structure is especially suitable for larger scale VPP, where different energy sectors are geographically distributed within different regions. In particular, comprehensive control is classified into two control levels, centralized and distributed control. In the first level, the centralized control, the VPP is centralized controlled to coordinate different distributed units. While different from the conventional centralized controller, the computational burden and storage requirement is distributed in the second control level. In the secondary control level, the distributed control realizes the local optimization based on local and neighboring information.

With the control method, the computation burden and storage requirement is relieved from level 1 centralized control to level 2 distributed control. Moreover, the hybrid control method also leverages the characteristics of centralized control structure, which can coordinate different units centrally and avoid disorders. However, due to the communication required between the two levels, the control method is also susceptible to cyber-attacks.

The different control structures for the VPPs are compared Table 1

5. Energy trading and economical aspects

Each of these four control models for VPPs has economic advantages and disadvantages. Centralized control methods offer greater command power compared to Decentralized approaches, leading to increased efficiency and cost savings by optimizing the dispatch of energy resources. VPPs using centralized control methods can also increase revenue through participation in energy markets and selling surplus energy to the grid. However, this approach has some drawbacks, mainly because it is susceptible to a single point of failure, thus if the central control fails, the entire system is at risk. Additionally, scalability is limited, making it difficult to handle a large number of DERs.

Distributed control methods, on the other hand, can handle multiple DERs, respond more quickly to changes in energy demand and renewable energy availability, and provide redundancy, if one component fails, the others can continue operating. However, there are also challenges with this type of configuration. Efficiency may suffer, as relinquishing central control means the system cannot fully optimize all resources. Moreover, maintaining multiple control systems incurs higher costs due to the need for multiple systems to operate simultaneously.

Hybrid control methods offer a balance between centralized and distributed control methods, combining the efficiency of the former and the scalability of the latter. They also enhance redundancy and reliability through the use of multiple control systems and provide greater flexibility to adapt to changing network conditions and energy markets. However, hybrid control methods require specialized interfaces for complex coordination and communication between control systems, and may not be as effective as centralized control methods for small or medium-sized VPPs with fewer consumers. On the other hand, decentralized control methods reduce implementation costs, enable prosumers to make decisions freely, and facilitate local energy markets and peer-to-peer energy trading. Additionally, decentralized control methods are less vulnerable to cyberattacks and other threats since there is no centralized control responsible for most of the management.

As can be seen, aggregators are very prominent for their complexity in different aspects, and although each of their models has economic advantages and disadvantages, this type of grouping will increasingly exist due to the different efficient solutions adapted to new smart grids of the future. Nevertheless, the insertion of this new structure represents a challenge for most countries, since the existence of an already established market, in general, opposes the presence of these new participants [150]. This section not only covers this context but also brings analyzes the tariff issue related to VPPs, their role, and what to expect in the future with the prices of the technologies involved falling. Other issues regarding electricity markets will be developed in Section 6.

5.1. Bidding analysis

With the rapid development of smart grid technology, utility users have become key participants in setting electricity prices through their involvement in demand response (DR) procedures. Smart grid technology enables the precise collection and analysis of historical information about users' consumption preferences. DR applications are an effective method to ensure energy balance in the electricity market, helping to mitigate the side effects of power fluctuations caused by the intermittent nature of renewable energy sources while maintaining user demand satisfaction. However, current DR methods primarily focus on demand-side management of the power grid. By incorporating DR into the smart grid framework, the ISO can dynamically communicate updated electricity prices to users, laying a foundation for intelligent management of users' utility consumption patterns and reducing costs.

Bidding analysis refers to the process of developing, evaluating, and optimizing strategies for submitting bids in electricity markets. For VPPs, this involves predicting future electricity prices, estimating the production capabilities of various energy sources, and assessing demand forecasts to make informed decisions about how much power to offer and at what price [151]. Effective bidding strategies are crucial for VPPs to maximize their profits, secure favorable contracts, and mitigate risks associated with market volatility. This analysis typically incorporates advanced algorithms, economic models, and market simulations to navigate the complexities of dynamic pricing and regulatory environments. By continuously refining their bidding strategies, VPPs can better balance supply and demand, integrate renewable energy sources efficiently, and enhance their overall market performance.

VPPs can optimize their bids in electricity markets to maximize profits by considering factors such as demand forecasts, renewable energy production, and market prices [152]. Successful bidding strategies

Table 1
Comprehensive comparison for different control structures.

Control structures	Advantages	Disadvantages
Centralized control	Global optimal solutions with coordination among subsystems.	(1) Computational burden with the increasing number of DERs; (2) Privacy. Independent privacy for different stakeholders.
Decentralized control	(1) Relieved computational burden and storage requirement; (2) Simplicity of control; (3) Independent of digital communication technology	Limited performance due to lack of information from other units.
Distributed control	(1) Distribution of computational burden to local processors; (2) Neighbor based communication eliminates the single point of failure; (3) Plug-and-play, scalability.	(1) Communication network is strongly connected; (2) Susceptible to cyber-attack; 3) Coordination issue among different subsystems.
Comprehensive control	(1) Fully control of different units; (2) Relieved computational burden and storage requirement.	(1) Susceptible to cyber-attacks; (2) Compatibility constrained by the communication bandwidth.

are essential to secure profitable contracts and avoid financial losses. Bidding strategies play a crucial role in the optimized operation and profitability of VPPs [153].

In [154], a comprehensive review of VPP operations is provided, focusing on resource coordination and multidimensional interaction. The study emphasizes an interdisciplinary approach, utilizing literature reviews and technical statistics to capture decision-making contributions to VPP operations.

Other studies, such as [155], address the challenges of bidding for VPP-enabled ancillary services in Australia's National Electricity Market (NEM). This study proposes a risk-averse optimal bidding strategy to handle the high randomness of system frequency deviation and significant price fluctuations due to renewable energy uncertainty. The authors developed a payment recovery mechanism designed to recover costs from frequency regulation (FR) undersupply. Mathematical tools such as Nash Equilibrium and the Distributed Best Response Algorithm are used to calculate the risk-averse optimal bidding strategy.

In [156], the authors propose an optimal bidding strategy and profit allocation method for shared energy storage-assisted VPPs in joint energy and regulation markets. They introduce a two-part price-based leasing mechanism that accounts for market prices and battery degradation, providing short-term use rights of shared energy storage to VPPs. Wozabal and Rameseder in [157] develop a multi-stage stochastic programming approach to optimize the bidding strategy of a VPP operating on the Spanish spot market for electricity. The VPP markets electricity produced in wind parks on the day-ahead market and on six staggered auction-based intraday markets. Their model, structured as a Markov decision process (MDP) and solved using a variant of the stochastic dual dynamic programming algorithm, demonstrates superior performance compared to deterministic models. In [158], Yang et al. propose an optimal bidding strategy model for a renewable-based VPP in the day-ahead market (DAM) that encompasses energy, reserve, and regulation markets. [159] proposes a day-ahead robust bidding strategy for multi-energy virtual power plants (MEVPPs) in the peak-regulation ancillary service market, addressing uncertainties in energy sources and loads. They establish a two-stage robust bidding model aimed at minimizing operation costs and analyze the influence of these uncertainties on the peak-regulation market.

It is important to note that bidding analysis is a critical component in the operation of VPPs, enabling them to effectively navigate the complexities of electricity markets. By leveraging advanced algorithms and interdisciplinary approaches, VPPs can optimize their bidding strategies to maximize profits, secure advantageous contracts, and mitigate financial risks. The studies reviewed highlight the importance of integrating various energy sources, managing uncertainties, and employing robust optimization techniques. As smart grid technology evolves, the continuous refinement of these strategies will be essential for VPPs to maintain a reliable balance between supply and demand, enhance market performance, and achieve long-term sustainability. The next section will delve into the profit analysis of VPPs, examining how different revenue models and operational strategies impact their overall profitability.

5.2. VPP profit analysis

The profit analysis of VPPs involves a comprehensive evaluation of operational costs versus the revenues generated from various streams such as energy sales and ancillary services. Understanding the different revenue models is crucial for optimizing profitability and ensuring sustainable operations.

One of the primary revenue streams for VPPs is energy sales, which involve selling generated electricity in wholesale markets. The pricing in these markets is highly dynamic and influenced by factors such as demand forecasts, market competition, and the integration of renewable energy sources [160]. VPPs that can accurately predict market prices and strategically allocate their generation capacity are more likely to secure higher profits.

In addition to energy sales, VPPs can benefit from renewable energy incentives. Governments and regulatory bodies often provide financial incentives to encourage the adoption and integration of renewable energy sources [161]. These incentives can take various forms, including tax credits, feed-in tariffs, and renewable energy certificates. By leveraging these incentives, VPPs can offset some of their operational costs and enhance their profitability. Moreover, the participation in green energy programs not only improves the financial performance of VPPs but also contributes to environmental sustainability goals.

The energy markets in which VPPs operate also significantly impact their profitability. Different markets have varying rules, price structures, and levels of competition. VPPs need to adapt their bidding strategies and operational plans based on the specific characteristics of each market [55]. Participating in multiple markets, such as day-ahead, intraday, and ancillary services markets, can provide VPPs with more opportunities to generate revenue. However, it also requires sophisticated market analysis and flexible operational capabilities to respond to market signals effectively.

The profitability of VPPs is influenced by a combination of revenue models, operational strategies, and market conditions. By optimizing energy sales, leveraging renewable energy incentives, and participating in demand response programs, VPPs can enhance their revenue streams. Additionally, strategic operational decisions and market adaptability are crucial for maximizing profitability. The next section will provide an in-depth look at specific case studies and real-world applications of these principles in VPP operations.

[55] provide a comprehensive review of various VPP models and their integration into electricity markets. Their analysis covers operational strategies, market participation, and technological advancements, highlighting the challenges and opportunities associated with VPPs in modern electricity markets. They emphasize the importance of innovative bidding strategies and market mechanisms in enhancing VPP profitability and reliability.

[156] present an optimal dispatch model for VPPs that incorporates carbon trading and green certificate trading mechanisms. This model aims to improve the economic and environmental efficiency of VPP operations by integrating wind power generation, photovoltaic power generation, gas turbines, and energy storage devices. Their work underscores the significance of integrating diverse energy sources to optimize VPP performance.

Other important study is [162]. It develop a windfall profit-aware stochastic scheduling model for industrial VPPs, incorporating both risk-seeking and risk-averse preferences. Their model includes best- and worst-case scenarios, quantified by the value at best (VaB) and the conditional value at risk (CVaR), providing a robust framework for decision-making in uncertain market conditions.

Additionally, [156] propose an optimal bidding strategy and profit allocation method for shared energy storage-assisted VPPs in joint energy and regulation markets. Their study introduces a two-part price-based leasing mechanism that considers market prices and battery degradation, offering a novel approach to managing short-term use rights of energy storage for VPPs.

A more general view is provided by [163], which evaluate the economic benefits of VPPs for both demand and plant sides using cooperative game theory. Their study focuses on a VPP composed of photovoltaic (PV) and energy storage systems (ESS) in an urban area of Japan, demonstrating the potential economic advantages of VPPs in urban settings .

The profitability analysis of VPPs underscores the importance of strategic management and market adaptability. VPPs that leverage dynamic energy markets, renewable incentives, and sophisticated operational strategies are well-positioned to maximize their profits while contributing to sustainable energy goals. The integration of innovative bidding strategies, market participation models, and the application of advanced technologies such as energy storage and demand response programs further enhance their operational efficiency and revenue potential. Future research and case studies will continue to shed light on best practices and innovative approaches that drive the profitability and sustainability of VPPs in an evolving energy landscape.

5.3. Relationship between Virtual Power Plant, Conventional Power Plan and government

Currently, in countries, there is a typical structure, usually dependent on carbonaceous sources, that provides electricity to their populations. It can be represented by these 3 phases: generation, transmission, and distribution of energy. In general, several companies have established themselves in one of these stages of production and from which they derive their profits. All this scenario presented can be called Conventional Power Plant(CPP).

On the other hand, there is a tendency for the emergence of prosumers, who will seek to gather in aggregators. In this way, as seen earlier, they will use energy more efficiently, sell the surplus and provide ancillary services. That will create a scenario in which owners of DERs will not only be less dependent on the CPP but will also compete with it in providing energy [150,164].

A lot of texts ignore this trend that will be gradually forming and focus only on maximizing the profits and advantages of the VPPs. However, many authors, such as in [163], consider this perspective a little unrealistic, because aggregators will continue to depend on the energy supplied by traditional companies, only on a smaller scale. And at the moment these companies decrease their profits, they will likely raise their prices to offset the expenses. Soon, a dispute of interests, in which the CPP has less demand for its energy and the VPPs have to pay more for the electricity supply, will begin to form. Many articles try to find the solution to this based on game theory.

5.3.1. Game theory

Game theory is a mathematical tool that analyzes models where two or more rational agents interact with each other and the action of one party interferes with the results of the others, generally, it is useful to describe economic relations in the most varied areas [165].

Although discussions involving game theory have existed since the 16th century, it was not until the 20th century with John von Neumann's "On the Theory of Games of Strategy" [166] that it began to be seen as a unique and important field of knowledge. From then on, it grew exponentially and was applied to a wide range of behavioral relations in society. Some examples of game types are Cooperative/non-cooperative, Simultaneous/sequential, Bertrand Competition, Zero-sum/non-zero-sum, Cournot Competition, Bayesian game, and others.

5.3.2. Approaches on Virtual Power Plants

As many have observed, the relationship between CPP and VPP may be harmful to both parties because each prioritizes their interests. To find mutually beneficial solutions, game models are often utilized. Furthermore, many of these schemes require an active government to ensure the parties' interests are protected, because policymakers can act as organizers of contracts and rules to be followed by the players. Some solutions addressed in the texts are [167]:

- Cooperative game theory is a model in which all players agree to relinquish certain individual advantages and adhere to rules agreed upon by the group. This allows all parties to participate and benefit from the arrangement, making it an interesting solution for VPP, which seeks to balance the interests of all involved. Some articles highlight the importance of policymakers in achieving this balance through subsidies, constraints, laws, and other incentives or limitations [163,168,169]. This game model can also be applied on an internal scale within aggregator organizations [170]
- Multi-leader-follower game theory organizes parties hierarchically into two main categories: leaders at the upper level and followers at the lower level. In this case, Nash's non-cooperative competition plays a crucial role in establishing relationships between the parties. The theory is useful in selecting an optimal price for VPPs in scenarios where other sets and distributors aim to secure their interests [171]. Aggregators can use it to choose the most profitable contracts in a competitive distribution network. It can also be applied to the internal organization of a VPP [172]
- Potential game theory was introduced in 1996 by scientists Dov Monderer and Lloyd Shapley. In this game, the goal of all players can be expressed using a single global function called the potential function. In the case presented [173], an optimal dispatch strategy was modeled based on Potential Game theory (PG) for VPPs. It takes into account the presence of multiple agents and aims to select models that offer the best and most balanced returns to the global distribution network and each VPP.
- The model of Stackelberg's Game Theory involves a leader that moves first and competes in quantity with other organizations that move in sequence. In [174], the VPP and the market operator assumed these roles at different times. The paper proposed two scenarios: in the first, the market operator is the leader and decides market clearing prices while taking into account the power loss minimization of VPP. In the second, the VPP operator becomes the leader to facilitate demand-side management (DSM) through appropriate monetary compensation.

5.4. Tariffs

Before the widespread use of Distributed Energy Resources (DERs), it was much simpler to charge consumers for their energy usage. The energy was provided by the distributors, and tariffs were adjusted based on the time of consumption to encourage a more even distribution of demand throughout the day. This helped prevent situations where there was a high concentration of energy usage at a specific time, which could cause damage to the energy network.

Nevertheless, with new technologies emerging, these forms of the collection began to become obsolete, since they did not encompass a coherent way of dealing with the prosumer [175]. They, in general, took very little – or did not take into account – the presence of RES, DGs, ESSs, EVs, and smart meters. Then, new methods began to emerge not only to cover these technologies but also to encourage people to use them [176].

This incentive has been occurring because the vast majority of DERs are linked to renewable sources and, consequently, to decreasing greenhouse gas emissions, which is a major focus for countries around the world [177]. It is also linked to the help DERs can provide to the grid with services [178], as there has been an exponential increase in electricity demand, which the traditional-centralized network would most likely have difficulty dealing by its own [179].

Although all VPP models can benefit from a more comprehensive pricing model, tariff policies play a crucial role in the deployment of DVPPs and EV-VPPs, especially since these models rely entirely on renewable sources. By using tariffs that incentivize energy production, the high initial costs of infrastructure – which remain significant despite the decreasing prices of storage and generation systems – can be offset. Such measures are essential to ensure the viability and scalability of these innovative power solutions.

5.4.1. More comprehensive tariffs schemes

There are a few ways to implement beneficial tariffs for DERs. Some of them are Real-time pricing (RTP) or Dynamic pricing, Critical peak pricing (CPP), and Time-of-use pricing (ToU) [180].

- In the RTP system is designed such that prices vary by hour, with the wholesale market as a parameter, and values based on the concessionaire's real-time production costs. When combined with intelligent measurement methods, prosumers can arbitrate the use and sale of their energy. Typically, they sell their excess energy during peak hours, when the revenue is higher, and purchase energy during off-peak hours, when costs are lower. This strategy indirectly helps the network, particularly by reducing demand during critical hours [181,182].
- In the CPP, prices are significantly higher at some intervals of time throughout the day or year, as these are hours of high electricity use, demanding too much from the grid. However, a discount will be agreed in all other periods, and may be beneficial if the customers plan their energy consumption. This encourages the use of DERs, since on days of critical price prosumers can use their own energy and on cheaper days buy from distributors [183].
- In the ToU, the day is divided into several periods, in which different prices and fees are applied. The prosumers are free to organize themselves in a way that is most advantageous to them. This price organization reflects higher marginal production costs during peak production into the consumer tariffs. It is a very interesting scheme for DER owners for the flexibility and the profits [184].

5.4.2. Multi-period dynamic tariffs

Given these updates, some studies have proposed to find an ideal rate organization to encourage participants of VPPs. To achieve this goal, a lot of complex points need to be thought out and solved. It is reasonable, for example, to provide tariff advantages to prosumers who have invested in both a generation system and ESS, as they have made a greater investment and are contributing more to the VPP than someone who has only a generation system. However, this raises some questions such as: how to appropriately compensate different categories of prosumers and encourage their aggregation into VPPs?

Trying to solve this and other issues, the article "Multi-period Dynamic Tariffs for Prosumers Participating in Virtual Power Plants" [83] proposed an efficient and general model for tariffs. Using dynamic multi-period export tariffs based on the dual variables of the optimization problem, authors had favorable results in not only stimulating the participation of prosumers in VPP but also in benefiting the different spectrum of them- with those who made more expenses having the best rates. Although more studies need to be done, this type of result shows that with the correct tariffs model, it is possible to improve the creation of VPPs and correctly benefit the distinct categories of their participants.

5.5. Future costs of the technologies involved

Over the years, Renewable Energy Sources are becoming more popular and accessible. According to IRENA, for instance, the global weighted average LCOE, which represents the current net cost of generating electricity for a generator over its useful life and serves to guide investments, fell by 88% between 2010–2021 to utility-scale solar PV projects, 68% for onshore wind and 60% offshore wind [185].

Along with this cost reduction that is taking place in energy generation [186], it is also worth highlighting the reduction related to Energy Storage Systems [107,187]. The average cost of lithium-ion batteries, for example, which are rapidly becoming popular around the world and are widely used in various systems, e.g., microgrids, had a reduction of 89% in its pack between 2010–2021 [188,189], and everything indicates that this downward trend in prices will continue to occur in the coming years [190]. Such a change ends up affecting other technologies like electric vehicles because they also have been experiencing a drop in their costs [191]. This is all influenced by the price variation of carbonaceous sources as well, which are further stimulating renewable sources as explained in [192]. All these changes in technology prices lead studies to suggest that VPPs are getting closer to becoming a common element in electrical ecosystems worldwide.

5.6. Analysis of different categories of VPPs from an economic perspective

Throughout this review, various categories of VPPs have been presented, each possessing unique characteristics that differentiate them from one another. As a result, the economic role of each class varies significantly depending on its main features.

As noted previously, CVPPs prioritize economic optimization and are therefore an attractive option for maximizing efficiency, reducing costs, and generating revenue from energy sales. Typically, these plants are operated by commercial entities and are best suited for tariffs that incentive renewable energy generation, as the focus is on the financial optimization of DERs. Due to this emphasis, centralized structures have been favored in the current context, often established through contractual agreements with prosumers, whereby the CVPP operator assumes control over DERs.

On the other hand, even with the primary objective of providing network adequacy and ancillary services, TVPPs offer several economic advantages, particularly in terms of cost reduction by utilizing existing DERs. This approach reduces the need for investments in network infrastructure upgrades, offering greater flexibility through the use of advanced algorithms that can be adapted and optimized according to network needs. TVPPs are also highly scalable, allowing them to be easily expanded or reduced by adding or removing DERs from the network, without the need for significant network investments. However, there are also some disadvantages to TVPPs such as the high costs involved in the deployment of advanced technologies and software, particularly when compared to CVPPs. As the primary profits of TVPPs are linked to revenue generation through the provision of network adequacy and ancillary services, they have fewer sales options than CVPPs would have if they were actively participating in electricity markets.

In terms of other categories, EV-VPPs offer cost savings for both EV owners and the grid. EV owners benefit from revenue generated by the power supply, while the grid benefits from the storage capacity that helps to reduce peak demand and provide ancillary services. Moreover, this type of VPP represents a zero-carbon energy storage model that can incentive greater use of renewable generation. DVPPs are also models that will become increasingly common, as most prosumers have renewable solutions in their homes, photovoltaic and battery systems. The lower costs of these technologies, along with more comprehensive tariff systems, are allowing DVPPs to develop and gain increasing importance in electrical networks. Subsidies from governments and institutions through cooperative schemes have played an important role in the development of these VPPs. Furthermore, they represent a more tangible solution for prosumers than local energy markets and peer-to-peer energy trading, as they are organized systems that collect various information for process optimization, unlike those that are more dispersed and less efficient solutions.

Fog-based Virtual Power Plants are a recent technological advancement that offers several economic benefits. By leveraging computing resources closer to the edge of the network, these systems can significantly reduce latencies in data processing and decision-making, making energy management more efficient. Additionally, they enhance scalability, particularly in regions experiencing rapid energy demand growth, where traditional grid infrastructure may struggle to keep up. FaVPPs also increase local control, empowering communities and neighborhoods with greater autonomy. Finally, because fog-based systems rely on local information stores, they tend to be more secure than other types of systems, which is particularly important for VPPs that handle sensitive customer information.

VPP 0.0 helps to reduce peak demand and smooth out grid fluctuations, which can lead to lower electricity prices for consumers and is primarily focused on power trading and balancing, however, it does not actively manage DERs in real time, which is less efficient and adaptable than other models.VPP 1.0, on the other hand, seeks more flexibility and responsiveness than VPP 0.0, as it includes demand-response capabilities and can adjust power usage in response to grid conditions and market prices. This can lead to greater cost savings for consumers and utilities, as well as more efficient use of renewable sources. Typically, VPP 2.0 is focused on economic optimization and therefore uses more advanced technologies such as artificial intelligence and machine learning, which can lead to further cost savings and efficiency gains. able energy sources. As said, VPP 3.0 takes into account the problem of optimal power flow, which seeks to determine the ideal operating points of generators, transformers, and other devices in the electrical system to minimize the total cost of generation and transmission, having as a limitation a series of restrictions. Through real-time management, it manages to optimize production, reducing costs and increasing revenues.

Compared to VPPs, P2P energy markets have limited economic organization. While P2P energy trading platforms facilitate direct transactions between individual producers and consumers, eliminating intermediaries, they lack the same level of resource optimization and coordination found in VPPs due to their decentralized nature. This decentralized approach can, many times, lead to inefficient energy utilization and unnecessary losses, potentially resulting in financial setbacks. It is also important to note that VPPs have the advantage of aggregating diverse energy resources, enabling their active participation in various markets with substantial power. In contrast, P2P systems are constrained to direct exchanges between participants, limiting their market reach. Furthermore, VPPs play a crucial role in optimizing the intermittency of energy resources and providing valuable services to the grid. This helps mitigate losses caused by the unpredictable nature of renewable energy sources and reduces the need for significant investments in infrastructure.

VPPs offer similar advantages to those mentioned earlier for P2P systems when compared to LEMs. However, LEMs have distinct economic benefits that deserve attention. They enable a reduction in transmission and distribution costs by generating and consuming energy locally, thereby minimizing the requirement for extensive long-distance infrastructure for transmission and distribution. Furthermore, LEMs promote community involvement and empowerment, as local participants can directly engage in energy transactions within their neighborhoods or communities. This active participation often leads to local economic development and creates opportunities for small-scale energy producers, thus generating new business prospects. As such, there is potential for growth for energy service companies, innovative energy startups, and even aggregators. These entities can provide energy management services, initiate local energy projects, and provide community employment opportunities.

6. Operation of VPP

In addition to their economic benefits, VPPs can participate in various ancillary services and electricity markets, providing reliable energy to the network. By using game models, tariff schemes, and advanced technologies, VPP clusters can maximize profits from the energy system while achieving various operational goals, such as energy balancing or providing ancillary services. Depending on the specific planning problem, different formulations can be used to optimize VPP operation. For example, one popular application of VPPs is providing grid services, where various units of the VPP coordinate to meet demand.

Before describing them, it is important to emphasize the importance of price signals in the electricity sector. They play a crucial role in conveying the value of electricity at any given time. These signals are determined by the interplay of supply and demand, which can fluctuate rapidly depending on various factors, such as weather conditions, time of day, and overall system demand. Usually, prices in electricity markets are specified by the marginal cost of producing the next unit of electricity, which is based on the cost of the most expensive generator required to meet demand. They are also influential in encouraging the development of new generation resources, including renewable energy and VPPs, and in provide economic incentives for investment, which can facilitate the transition to cleaner energy sources. Price signals help market participants make informed decisions about production and consumption by reflecting the cost of producing electricity at any given moment. This, in turn, fosters competition and innovation, driving down costs and promoting efficiency in the electricity sector.

In addition, it is also important to highlight VPPs applicability within the challenges of operation and planning of energy generation and transmission. It is even plausible to say that such virtual energy solutions will likely become common practice soon. Two important works with this vision are [193,194]

Overall, the first is a book that provides an introduction to VPPs and electricity markets, with a particular focus on the challenges of decision-making under uncertain conditions. It thoroughly examines the key components of VPPs and their role in smart grids, shedding light on scalability, neutrality, and risk aversion issues. Furthermore, the book investigates the involvement of VPPs in energy and reserve markets, addresses the complexities of dealing with uncertainties, and presents optimization methods to mitigate them. It also explores the ability of VPPs to influence market prices and examines the complexities associated with planning the expansion of VPPs in the generation and transmission sectors demonstrating how VPPs will be a solution for the future in these fields.

The latter refers to a doctoral thesis that focuses on the optimized operation and planning of VPPs in electricity markets. This thesis provides a precise model for various components of VPPs, including both conventional and renewable generators, storage systems, and demand management. It takes into account the uncertainties associated with RES, such as fluctuating levels of renewable generation, production costs, and market prices. To address these uncertainties, the author introduces two operational models. The first model deals with the supply strategy in the day-ahead (DA) energy market, considering the VPP's participation in the real-time energy market to compensate for power deviations. The second model focuses on the day-ahead self-scheduling problem of a VPP in energy and reserve electricity markets. It considers uncertain reserve deployment requests and incorporates confidence bounds and scenarios to effectively model uncertainty. The work presents several significant findings. For instance, it demonstrates that the Adaptive Robust Optimization stochastic approach is well-suited for handling uncertainties in the supply strategy of a VPP in the Day-Ahead Market (DAM) energy market. Compared to deterministic approaches, this stochastic approach enhances profits by effectively addressing uncertainties. Additionally, it highlights the importance of demand flexibility and enables arbitrage opportunities in different electricity markets. These studies are extremely important, as they open doors for the insertion of VPPs in these markets.

6.1. Operational steps of a VPP

The operation of a VPP involves the coordination and management of a variety of DERs, such as renewable generators, energy storage systems, and demand response programs, to supply energy to the electrical system in an optimized and flexible manner. The VPP acts as a virtual hub that integrates these decentralized resources, enabling them to function as a single cohesive entity in the energy market.

1. Monitoring and Control: The operation of the VPP begins with the continuous monitoring of connected DERs. Sensors and real-time communication systems are used to collect data on energy production, state of charge of storage systems, energy consumption, and market conditions. This data is processed by advanced control systems that coordinate the operations of each DER.

2. Optimization of Production and Consumption: Based on the collected information, the VPP utilizes optimization algorithms to determine the best way to dispatch energy. This may include deciding when to store or release energy from storage systems, when to adjust the production of renewable generators, and when to activate demand response programs to balance supply and demand.

3. Market Participation: The VPP actively participates in energy markets, such as the Day-Ahead Market (DAM), Intraday Market (IDM), and Real-Time Market (RTM). For each market, the VPP submits bids and offers based on the optimization of available resources, aiming to maximize profits while meeting energy needs and providing ancillary services to the grid.

4. Response to Fluctuations: One of the key advantages of the VPP is its ability to respond quickly to changes in grid conditions, such as fluctuations in demand or energy supply. The VPP can rapidly redistribute production among DERs or adjust energy consumption to maintain balance in the grid, ensuring the stability and reliability of the energy supply.

5. Coordination of Ancillary Services: In addition to energy production, the VPP can also provide a variety of ancillary services, such as frequency regulation, voltage control, and operating reserves. The flexibility of DERs allows the VPP to offer these services efficiently, enhancing the resilience and efficiency of the electrical grid.

6. Management of Uncertainties: The operation of the VPP is continuously adjusted to deal with uncertainties, such as the variability of renewable generation and market price fluctuations. Stochastic models and robust optimization strategies are often employed to mitigate the risks associated with these uncertainties, enabling the VPP to maintain stable and profitable operations.

These processes make VPPs an innovative solution for integrating distributed resources into the electrical grid, promoting the transition to a more sustainable and resilient energy system.

6.2. Day Ahead Market

One of the most known models of these markets on electricity is the day-ahead market. It corresponds to a financial electricity market where participants can buy and sell electricity for each hour of the next day [195]. DAM is a forward market, i.e., it means that electricity transactions are made in advance, allowing participants to present offers to sell or bid to buy electricity for each hour of the following day. Prices in this model are intrinsically determined by the relation between supply and demand, which, in turn, are influenced by climate, electricity availability, and production. Many institutions can participate in this configuration as power generators, transmission system operators, electricity suppliers, large consumers, and VPPs. DAM presents several advantages like price transparency, competition, efficient resource planning, reduced market volatility, stimulation of renewable energy sources, and risk management due to the price signals.

- Market participants submit their offers to buy and sell electricity to the energy exchange. They include the amount of electricity to be negotiated and its price
- The power exchange sorts the bids and offers based on price/quantity and organize them
- The power exchange calculates the price at which the amount of electricity demanded by buyers equals the amount of electricity supplied by sellers, it is called clearing price
- it is determined which bids and offers will be accepted based on the clearing price
- after this sequence of steps, the delivery of electricity to buyers is organized.

Given this context, The ease of entry and participation in virtual power plants (VPPs) has led to an increase in studies focusing on bidding strategies. In [158], for instance, an optimal strategy considering a great presence of renewable sources associated with a mixed-integer nonlinear programming model to reach the best profits of the VPP in various cases proved to be effective. In [196], a probabilistic model for optimal day-ahead scheduling of electrical and thermal energy resources in a VPP is developed. It takes into account the participation of energy storage systems and demand response programs (DRPs). The model schedules energy and reserves simultaneously, considering the uncertainties of market prices, electrical demand, and intermittent renewable power generation, using the Point Estimate Method (PEM) to model uncertainties. In [197–200] different management models of VPPs based on renewable energy sources are presented as well. The Refs. [201–203] focused on maximizing the profits of the VPPs participating in DAM. It is important to emphasize that due to their size, aggregators can play the role of buyers and decision-makers as well.

6.3. Intraday Market

In the Intraday Market (IDM) electricity exchanges are characterized by same-day trading. This means that any positions opened during the day must be closed by the end of it. IDM enables participants to adjust their positions in real time, usually within the day-ahead timeframe. It also enables them to buy or sell electricity for delivery within the same or on the following day. It is important to emphasize that the main difference between this market and DAM is the timeframe for trading and settlement, as the first allows to participants adjust their positions in real-time, while the second only allows participants to negotiate energy to be delivered in the next day. IDM also makes it possible to match energy exchanges with higher accuracy and also to correct imbalances in real-time due to the unpredictability of renewable energies. Various studies have examined the use of IDM as a means for VPPs to sell their energy and boost their profits since VPPs offer a flexible, cost-effective, reliable, and sustainable option for this market option. The potential benefits of VPPs participating in IDM have been investigated by the following [157,204–208].

6.4. Real-time Market

In Real Time Market (RTM), trading takes place in or near real-time, typically within a period of minutes to hours, and this immediate horizon for trading is the main difference between the previously presented models. It is designed to balance supply and demand on a minute-by-minute basis and ensures that the electrical grid remains stable and reliable. Prices are typically determined by a market matching algorithm that matches available supply with demand and is influenced by the cost of generating electricity, transmission constraints, availability of reserves, etc. Based on this, participants can buy or sell electricity at a variable price at a given time. Because RTM usually works together with the DAM and represents a mechanism to instantly balance supply and demand, VPPs have shown promise in participating in these markets, especially due to their ability to respond quickly to changes in grid conditions. A lot of articles focus on modeling and increasing the profits of VPPs, some recent examples are [156,172,198,209–211].

6.5. Bilateral contracts

A Bilateral Contract is an accordance between an energy producer and consumer where they agree on terms such as the price, volume of energy, contract period, and minimum power supply/consumption. These contracts are beneficial, since they provide both parties with price certainty and stability, as the agreed price for the electricity is fixed for the duration of the contract. Bilateral contracts also enable electricity generators to secure long-term sales agreements, which can help them finance new power plants or renewable energy projects, which are essentially related to VPPs. As it is a more decentralized alternative, it allows for greater flexibility than other types of electricity contracts. VPPs can increase the value of bilateral contracts mostly by offering a more reliable source of electricity since it aggregates and optimize DERs and provide a predictable and controllable source of electricity, which can help reduce the risk of supply interruptions or price volatility [212–215].

6.6. Energy balancing market

An Energy Balancing Market is a model of the electricity market that helps balance supply and demand in or near real-time, playing, because of that, a critical role in ensuring the reliability and stability of the grid. In the market, grid operators use a variety of mechanisms to ensure that sufficient electricity is available to meet demand at all times such as demand response, which represents the incentive of electricity consumers to reduce their demand during times of high electricity use and balancing mechanisms, that enables operators to make adjustments to the amount of electricity being generated or consumed in response to changes. This market is essentially important, due to the highly variable and difficult-to-find demand patterns, especially as more renewable energy and probabilistic sources become more accessible. VPPs are important in energy balancing markets because they can help maintain the supply and demand of electricity on the grid stable in real-time, mainly due to the flexibility provided by controllable resources of consumption and generation [201,216–218].

6.7. Ancillary service market

Ancillary Services (AS) are essential services required by transmission and distribution system operators to maintain the integrity and stability of the power grid, as well as power quality. In the past, when the grid was predominantly unidirectional, centralized, and with a top-down structure, the system operator had more control to maintain balance and function with the use of AS. However, the growth of decentralization with DERs and increased demand with electrification has presented challenges for operators worldwide, as they were not prepared for such changes in such a short amount of time. This led some grids to adopt restrictions to maintain a proper function, due to limited decentralized generation. Puerto Rico Electric Power (PREPA), for example, narrows RESS fluctuations to less than 10% (of nominal power)/per minute. Mexico took a similar action, except for 5% (of nominal power)/per minute [219]. And, more recently, discussions took place in Australia on whether households should pay for the right to export more solar energy to the grid.

Such constraints are an example of the severe actions grid operators are forced to take upon rapid scenario changes. However, coordinating distributed resources has been shown to bring benefits and reduce the requirements for such severe constraints, and VPPs in particular are known to be particularly beneficial since they can represent a large amount of flexibility that can be easily coordinated and dispatched [220,221].

Next, we discuss in more detail each of the major AS, and correlate their fundamentals with operations combined with VPPs.

6.7.1. Frequency services

Frequency services have the function of stabilizing the balance between energy supply and demand by keeping the system operating at levels close to the nominal. Because of that, there is a constant need to keep the network frequency running properly, and ancillary services can do that by helping the network respond quickly when fluctuations or changes occur that would impede the supply of electricity.

It is important to mention that the names given to the operations may be different depending on the country. In the USA, for example, services can be divided into regulation, spinning reserve, non-spinning reserve, and replacement reserve, while in Europe, which will be used to describe the services, are divided into 3 categories: primary reserve, secondary reserve, and tertiary reserve.

- Primary frequency response (PFR), also called primary control reserve and frequency responsive reserve corresponds to the first stage of frequency control in a system. It restores the balance between generation and consumption within seconds of disturbance occurring and is an automatic response to locally correct deviations being triggered automatically until the frequency reaches a steady.
- secondary reserve, also known as automatic Frequency Restoration Reserve (aFRR), fix area control error of the primary frequency control maintaining the desired energy exchange of it with the rest of the synchronous area. When there is an imbalance of production and consumption, the aFRR is activated by the central controller in order to solve these problems. Secondary control is activated after a few seconds and usually ends after about 15 min of operation- If the cause of the control deviation is not eliminated after that time, the secondary control will be handed over to the tertiary control.
- The tertiary control reserve, called also manual Frequency Restoration Reserve (mFRR), is mainly needed to adjust large and long-lasting control deviations, especially after production stoppage or unexpected and load changes and, it is activated manually. The mFRR replaces secondary control reserves with the aim of restoring a sufficient secondary control band. Despite being longer than the previous two, it aims to restore normality and the correct functioning of the system in minutes.

It is important to mention that VPP represents a great opportunity to participate in the frequency regulation market (FRM) because the presence and organization of DERs allow quick responses to deviations [96]. This is shown in [222], in which an optimal bidding strategy of a VPP participating in the day-ahead FRM and the energy market (EM) is proposed. This paper also has shown good results for VPP profits and in the provision of services, especially aggregators with diversified generation resources. In other research, such as [223,224] it is possible to also find coordinated control and forecasting strategies for VPPs in the provision of frequency services in a DVPP. Other categories of VPPs, like EV-VPPs [94], have shown good results in providing this type of service. It can be noted that DERs by themselves are great in the provision of frequency services, due to their speed of response, however, these network aids can be greatly enhanced if these resources are arranged in a VPP.

6.7.2. Reactive power and voltage control

Reactive power services make sure voltage levels on the system remain within a range, above or below nominal levels, that guarantee a proper operation of consumer loads. As seen, the penetration of distributed resources brought new problems, such as the increase in peak voltage. Still, if the reactive power capability of DERs is properly explored, this and other problems can be solved.

For this arrangement to work, however, some obstacles need to be overcome. One of the issues consists of the fact that the capacitive behavior of the distribution system impairs the voltage stability of the transmission system, and, for this reason, it is necessary to monitor the exchange of reactive power between DERs and TS very well. Another is related to the additional costs for DERs owners due to the prolonged use of reactive power. Some problems are related to the direct connection to DS instead of transmission, such as the imbalances presented by the DS, the greater harmonic pollution of voltage/current, and the fact that some DERS can also act as harmonic filters [219]. Therefore, so that the voltage regulation in the distribution network and the exchange with the transmission system work properly in the presence of DERs, the reactive power of the DERs must be controlled in an orderly manner. And one of the main ways to do this is with the implementation of VPPs.

Some articles attempt to investigate these deployments. In [225], the authors investigate the optimal active and reactive power scheduling of PHEVs and DGs in VPPs. In [226], it presents a schedule for managing these variables in markets through an arbitrage strategy for VPPs using a mathematical model with constraints and solved by MILP, which showed good results in aggregate profits and service provision. As seen before, great control of the DERs is necessary for the reduction of voltage-related failures. Because of this, the paper [227] investigates 4 scenarios of energy management systems through the integration and communication of Hydroelectric Power Plants in a VPP.

6.7.3. Black-start services

These services are requested when a system-wide blackout occurs. They re-energize the transmission system and provide start-up power to generators that cannot do it on their own. Generally, they depended on large thermal plants connected to the TS and energized the entire system from a “top-down” structure. However, with the increase in the decentralization of networks, new approaches are being discussed, highlighting the “bottom-up” black start approach, which is analyzed in [228], in which the authors had good perspectives and believe that it can provide an adequate black start capability. With this context in mind, it is believed that with proper coordination of DERs such as those provided by a VPP, these services would be even more viable [229].

6.7.4. Scheduling and dispatch

To maintain the reliability and balance of the electrical grid, it is essential to ensure that the energy generated in the system is equal to the energy consumed. This requires careful scheduling and dispatch. Typically controlled by the TSO, scheduling, and dispatch are services focused on coordinating the generation and transmission units. Scheduling involves determining the amount and type of ancillary services required to maintain grid stability and reliability over a specific period. This includes forecasting system demand, assessing the availability of ancillary service resources, and coordinating the delivery of these services from multiple providers. Dispatch refers to the actual execution of the scheduled ancillary services. It involves selecting the most appropriate provider based on factors such as cost, availability, and technical capabilities, as well as monitoring the real-time conditions of the power system and making any necessary adjustments to ensure grid stability and reliability. Together, they are extremely important, because they ensure the smooth and reliable operation of the power system by providing the necessary ancillary services to balance the system and maintain power quality

For years, grid operators have relied on centralized power plants to provide ancillary services. However, with the increasing penetration of DERs, it becomes more difficult to centrally manage the variability and intermittency of these resources [230]. VPPs allow for the aggregation and control of distributed resources providing a more reliable and cost-effective alternative to centralized power plants. Thus, VPPs are important in scheduling and dispatching services, because they provide flexible and efficient means of managing the variability and intermittency of distributed power resources. Some articles relate this integration with VPPs and DERs are [231–233].

6.7.5. Operating reserves

Operational reserves (OR) are a type of ancillary service on the electricity grid that provide an energy buffer to ensure that the balance between electricity generation and consumption is maintained in real-time. The electrical network requires a constant balance between the amount of electricity that is generated and consumed, and any unwanted variation can cause interruptions and damage to the network. Operating reserves are used to help maintain this balance, providing a reserve of energy that can be drawn upon to handle sudden changes in demand or supply.

As it corresponds to an energy dispatch as a response when the generation is not keeping up with the load OR can be divided into two major categories: Ten-minute reserves and Thirty-minute reserves [234]. Many studies have analyzed the impact of RES growth on operating reserves [235–237]. In [238,239], the authors discuss, for example, methods and integration of wind generation associated with OR.

As VPPs mitigate the negative effects of the growth of DERs and enhance the positive ones, it is also possible that they effectively participate in the Operating Reserves services. In the article [240], an optimal VPP energy management method focusing on optimal energy and operating reserve (OR) scheduling is presented. Although the research showed good results, it also recognized the need for more studies on this topic related to VPPs since they present a great opportunity for the electrical system.

6.8. Analysis of the different categories of VPPs from an ancillary service perspective

In general, the different categories of VPPs provide power reliability and flexibility, as they can provide fast and accurate responses to changes in supply and demand, monitor network conditions in real-time, except for VPP 0.0, and adjust the output to maintain network stability and prevent blackouts or power outages. While the different categories of VPPs overlap in some of these respects, they also have individual strengths in providing certain ancillary services.

CVPPs, for example, can offer ancillary services at a lower cost than traditional power plants because their main focus is economic optimization. They stand out especially in frequency regulation services since CVPPs are typically operated by commercial entities that focus on maximizing profits and reducing costs through cost-effective optimization of DERs. With this in mind, frequency stability in the grid is essential for them to avoid losses, forcing CVPPs to continuously balance electricity supply and demand in real-time to keep the grid frequency within a narrow range through adjustments in the output power and their loads. TVPPs, on the other hand, are primarily focused on providing network stability services, such as voltage control, reactive power compensation, and congestion management. They achieve this by controlling a variety of flexible loads and generators, such as electric vehicles, energy storage systems, and industrial equipment, to provide reactive power support to the grid. Despite being able to participate in several services, TVPPs stand out in providing voltage control services for the grid, because they usually rely on advanced control algorithms and real-time monitoring systems that allow them to respond quickly to changes in voltage levels. In addition, TVPPs can take advantage of the features of DERs, such as battery storage systems and inverters, to provide fast and accurate voltage control services.

The EV-VPP category stands out in demand response services, due to the high presence of ESSs with batteries. However, to offer this type of service to the network, the consumer must adapt his energy consumption to the conditions of the network. In general, to support the grid, EV-VPPs can charge during off-peak hours and discharge during peak hours. With this behavior, they can help to reduce strain on the grid during periods of high demand. DVPP, in turn, can be considered a broader category than EV-VPPs, as it comprises demand response technologies, energy storage, and renewable energy resources. Soon it also stands out in demand-side services for the same reasons as EV-VPPs. However, DVPPs also have other highlights such as frequency regulation, voltage support, and reactive power control, mainly because they respond quickly and efficiently to changes in electricity demand and supply using advanced algorithms and machine learning to manage a diverse range of flexible power resources. FaaVPP has few states related to auxiliary services, however, it is believed that it will be essential to provide fast and reliable grid stabilization services, such as those related to frequency, due to its advanced computational capacity and data processing resources. fog computing data, allowing them to analyze data from multiple sources in real-time and make quick decisions.

Virtual Power Plant (VPP) 0.0 is the most basic form of organization regarding its type classifications as it does not have any advanced control or optimization features. Therefore, it is more limited in providing many advanced ancillary services to the network. However, it can still promote several services standing out in terms of peak reduction for the network. This is because it can supply power during periods of high demand through generators. VPP 1.0 and 2.0, on the other hand, incorporate advanced control systems and communication technologies to optimize VPP performance. Therefore, these VPPs are able to offer a wider range of ancillary services to the network than VPP 0.0. However, its biggest highlight is the provision of frequency regulation service, due to advanced control systems that allow adjusting the output of generators in real time to help regulate the grid frequency. Finally, VPP 3.0 can be considered the most advanced version of this model of VPP categories as it incorporates the latest technological advances in control systems, communication networks, and artificial intelligence. As a result, VPP 3.0 is capable of deploying a wide range of ancillary services to the grid such as voltage regulation, demand response, black start, etc. Of these, the quick response, especially about frequency regulation (a matter of seconds), of this category due to the advanced control systems and optimization algorithms that allow adjusting the output of distributed energy resources (DERs) in real-time are the ones that most call attention.

P2P systems facilitate the direct exchange of energy between participants, limiting their involvement in the ancillary services market. This limitation arises mainly due to the decentralized nature of P2P systems, which lack the typical coordination and centralization found in traditional organizations. Thus, the effective incorporation of P2P systems in auxiliary services would require the implementation of appropriate coordination, communication and control mechanisms. This may involve adopting more centralized features or collaborating with other systems such as VPPs. In addition, current regulatory structures and market structures generally do not offer much space for P2P participation in these types of services, due to their unpredictability and difficulty in guaranteeing prior responses to facts. Despite these challenges, there is potential for P2P systems to indirectly contribute to network stability by integrating DERs, implementing some models of demand response programs, and incentivizing prosumers.

LEMs can present challenges similar to P2P systems; however, it is important to note that the extent of these challenges depends on factors such as project scope, regulatory frameworks and market rules in a given jurisdiction. LEMs have the potential to offer a wider range of ancillary services compared to P2P models, depending on their design. In some cases, LEMs and VPPs can complement each other and work together, as discussed in Section 1.1.4. Consequently, based on the adopted structure and market rules, LEMs can provide a variety of ancillary services, mainly at the local level. These services can include frequency regulation, voltage control, reactive power, operational reserve and much more.

7. Experiences and implementations around the world

Until this section, the paper has focused mainly on the theoretical aspects of VPPs. However, it is essential to point out that there are numerous VPP projects implemented or in progress. Although many of these implementations tend to use less complex and centralized models, it is important to recognize the importance of these applications in advancing the participation of these new organizations in electricity ecosystems. Throughout this section, it becomes evident that there is considerable overlap between the concepts, which shows the extent to which the theory has become complex and unclear over the years — an example of this lies in the fact that certain VPPs can be classified in various categories. This observation is coherent with the aforementioned idea that the topic is still emerging, with different models and theories being developed by different authors. Many of these practical implementations also bring new ideas for regulatory advances and how the market will adapt to this new bottom-up structure. These regulatory advances will be discussed, bringing up two important cases, Market model 3.0 and Redispatch 3.0 (see Table 2).

7.1. America

On the American continent, projects related to VPP are mostly concentrated in the United States. This happens since the US wholesale market was one of the first to encourage liberalization and competition in the electricity sector through the “Public Utilities Regulatory Policy Act” in 1978, which gave great autonomy to society [241]. Of the existing projects, 3 will be discussed: AMS California, GMP Project, and SunRun VPP.

The first deserves recognition because it was related to the decommissioning of a nuclear power plant in California in 2012 and it was implemented when the VPP concept was extremely restricted to academic circles. In the beginning, the government thought of investing in a thermoelectric plant, however, through a series of studies and discussions, a group of companies led by AMS managed to convince the abandonment of this initial idea by proposing and demonstrating that the creation of a VPP would bring a series of benefits to the network and the population [242].

Table 2
Example of VPPs around the world.

Trial name	Project partners	Location	Possible classifications	Size/Scale	Duration
SunRun VPP	Sunrun and ISO New England	US	CVPP	5000 PV-battery systems	2019–2022
GMP Project	Green Mountain Power (GMP) and Tesla	US	CVPP/VPP 0.0	2000 small scale-users	2018-ongoing
Olympic Peninsula Project	Pacific Northwest National Laboratory and US Government	US	TVPP	112 homes with flexible loads	2007
AMS	Advanced Microgrid Solutions and SunEdison	US	CVPP	Unknown	20
Advanced VPP grid integration	SA Power Network, Tesla and the Commonwealth Scientific and Industrial Research Organisation (CSIRO)	Australia	TVPP	Unknown	2020
Simply VPP	Simpy energy, SA Power Network, Greensync and the Australian Renewable Energy Agency (ARENA).	Australia	CVPP/VPP 0.0	More than 1000 PV-systems.	2020
South Australia Testa VPP	South Australia (SA) Government and Tesla	Australia	CVPP/VPP 0.0	Up to 50 000 PV-battery systems (up to 250~MW in aggregate)	2019–2022
VPP Demonstrations	AEMO, third party aggregators and VPP.	Australia	TVPP/VPP 1.0	3–5 participants	2019-ongoing
VPP Powerclub South Australia	Power Ledger and Powerclub	Australia	CVPP/VPP 2.0	Hundred of users with PV-battery systems	2019-ongoing
AGL Virtual Power Plant	AGL, Federal Government, ARENA, Sunverge	Australia	CVPP	1000 storage systems in residential homes	2018
Smartpool	Next Kraftwerke	Germany	CVPP	“Next Pool VPP” has more than 2900 energy production and consumption units	2015
EDSON Project	DTU CET, IBM, Risø DTU, Siemens, Dong Energy, Østkraft (Bornholm), Eurisco and Danish Energy Association	Denmark	EV-VPP/TVPP/DVPP	Unknown	2009–2012
Couperus Smart Grid project	Stedin Netbeheer, Itho Daalderdorp, SWY, Eneco Trade, TNO and IBM	Netherlands	TVPP	300 residential users with flexible loads	2012–2014
PowerMatching City	Energy research Centre of the Netherlands (ECN), KEMA, Humiq and Essent New Energy.	Netherlands	TVPP	25 resideltial users with flexible loads and PV systems.	2008–2014
Nice grid	Enedis, Netseenergy energy company, Dad'kin, Réseau de Transport d'Électricité (RTE), Watteco,Association pur la Recherche et le Developpement des Methodes et Processus Industriels (ARMINES), Saft, Alstom Grid France and Électricité de France (EDF)	France	TVPP	1500 small-scale users	2012–2015
FENIX	EDF, AREVA, idea, Scal Agent, IBERDROLA, RED ELÉCTRICA DE ESPANÃ, Gamesa, ZIV, labein, ISET, SIEMENS, KORONA, ECRO, vrije Universiteit amsterdam, ECN, PÖYRY, THE University of Manchester, nationalgrid, Imperiall College Lodon and EDF energy	Spain and UK	TVPP	Unknown	2005–2009
Shanghai Huangpu District VPP project	Unknown	China	CVPP	Unknown	2016
Sustainable open Innovation Initiative	Kyocera, the Kansai Electric Power Co., ENERES Co., KDDI Corporation and Tokyo Electric Power Company (TEPCO) Group	Japan	TVPP	Unknown	2018–2019
AutoGrid-ENERES VPP	AutoGrid and ENERES	Japan	CVPP	More than 10 000 assets	2020–2021

The second was led in 2021 by Vermont utility Green Mountain Power (GMP). It used the Tesla Powerwall batteries, Tesla Autobidder software, and photovoltaic panels. This project is important to mention since GMP claimed that it has already saved its customers around 3 million dollars in total per year, mainly by using a tariff scheme adapted to benefit the participation of prosumers. Lastly, SunRun VPP is worth mentioning, because the company found the first Residential Virtual Power Plant to participate in Wholesale Market and also this aggregator was implemented at a crucial economic moment, as Americans were experiencing a runaway increase in their energy costs, with skyrocketing inflation and heat waves that are worrying operators.

South America, however, did not show such investments in VPPs due to a series of factors. Mainly because it already has mostly a renewable and safe energy matrix when compared to the rest of the world, especially due to the diversity of climatic conditions that favor a clean generation, making investments in VPPs a non-immediate necessity. In 2022, Brazil, for example, presented around 80% of its electricity generation from renewable sources, and within that number, 56.8% came from hydraulics. Meanwhile, the rest of the world continued very dependent on carbon sources and had only about 30% of its electricity coming from clean sources [243]. It is also worth noting that many Latin American countries have more closed electricity markets. Precisely because of these different contexts and reliable sources, some countries did not show exponential growth in decentralizing measures like the US. However, it is still possible to see a growth of DERs, prosumers, and studies of VPPs in Latin America [244–248].

7.2. Oceania

This subsection will focus on Australia because it has become an example in the implementation of photovoltaic energy in the world. While in 2010 there were just 100,000 photovoltaic panels in the country, currently, there are millions of photovoltaic installations with a combined capacity of over 28.2 GW [249]. This exponential increase in renewable resources associated with the great freedom of the Australian electricity market has increased the formation of VPPs [250].

An interesting project in the country is the VPP Powerclub South Australia. In it, Power Ledger teamed up with energy provider Powerclub to create a VPP based on P2P and blockchain energy exchange technologies. This project stands out from the rest for being one of the first to propose a decentralized solution in practice and because of that together with the financial benefits of users, it can be included in consumer-focused VPP (VPP 2.0).

Another important aggregator is the South Australia Testa VPP [251] since Tesla is aiming to form the world's largest VPP. To participate, the interested party must have photovoltaic panels and Tesla's power wall system of lithium-ion batteries. A central intelligence will process and organize these DERs of all users at certain times, which is an essential characteristic to include this VPP within the Retailer VPP (VPP 0.0).

The last project to be mentioned is the VPP Demonstrations [252]. It is important because it is one of the steps in a broad process to demonstrate that DERs can be integrated into the National Electricity Market (NEM) of Australia, bringing benefits to consumers and system security. Despite being a small-scale test structure, it was enough to confirm the provision of power and network support services of aggregators, reinforce the need to review the existing regulatory arrangements for VPPs in the energy markets and Frequency Control Ancillary Services (FCAS), analyze the current cybersecurity measures and provide data of the integration to NEM. This VPP model can be embedded in the system-focused VPP (VPP 1.0).

7.3. Europe

Currently, Europe has been an example of investments in renewable energy and distributed generation [253]. Along with this, a large part of its population has been investing in Distributed Energy Resources. An example of this is a study conducted by CE Delftestimates that 83% of EU households could become prosumers by 2050 [254]. Therefore, due to this exponential increase in the decentralization of the electrical system and the growing of people participating in energy trading, VPP has been gaining strength as a way of organizing European energy.

One of the pioneering projects was the project FENIX [255–257]. Implemented in Spain and UK, it is important for being one of the first to perform in practice a subject that was previously intrinsically theoretical, and in addition, it used a series of terms that have become essential for the description of VPPs, such as Commercial Virtual Power Plants and Technical Virtual Power plants.

Another innovative VPP is the EDSON Project implemented on the islands of Bornholm, Denmark [258]. Their focus was on studying the integration of electric vehicles and wind generation into the grid, being, therefore, considered an intersection of EV-VPP, TVPP, and DVPP. Other 2 projects worth mentioning are "The POSITYF Project", which was designed by Spain, France, Switzerland, and Germany focusing on the creation of a DVPP, and the "Vibeco (Virtual Buildings Ecosystem)", which is a VPP service designed by the Finnish government and Siemens aiming to create a digital platform where properties and industrial companies that have DERs can participate in a sustainable energy environment.

7.4. Asia

In Asia, a country that has stood out in investments in electricity and, consequently, in VPPs is China. The first Chinese VPP practical platform was developed in North Hebei, in 2019. Based on the published reports, the load during summer reaches about 6 GW in the region, of which 10% is already responded with the help of an aggregator, equaling a 600 MW conventional power plant. In addition, the thermal electrical load demand is also supplied by the VPP in real-time, which is expected to generate 720 GW per hour and a reduction of 0.6365 million tons of carbon dioxide. Another important aggregator was constructed at Huangpu Distributed, Shanghai, in 2021. In it, the demand response using the flexible load of commercial buildings is explored. It is also worth mentioning that, in May 2022, a VPP supported by the State Grid of Shen Zheng achieve 0.274 income per kWh [259].

Japan has also been paying a lot of attention to VPPs and their studies. Since The Great East Japan Earthquake of March 2011 and due to its climatic conditions, the region started to invest more in renewable generation [260] and, for this reason, DERs and aggregators became essential, especially for the efficient use of electricity. Currently, two projects are important. The first is "AutoGrid-ENERES VPP", which start in 2020 and already has more than 10 000 assets, being the largest storage-virtual power plant (VPP) in the world by asset volume. The second is the "Sustainable open Innovation Initiative", which was implemented by Japan's Ministry of Economy, Trade and Industry (METI), demonstrating the importance of governments in the implementation of these new organizations of DERs.

Another country that deserves recognition for studies of decentralized electrical systems is India, due to its large size and number of inhabitants. With such proportions, it is difficult to obtain a network that is always consistent and strong, therefore, studies on VPPs are essential for the country. Some of them, that analyze the Indian context and the benefits provided by VPPs, are [261–266]

Despite all the studies shown, there are many others related to VPPs in Asia. In [267], the South Korean context is analyzed. In [268], a Battery Energy Storage System (BESS) designed for VPPs in Malaysia is considered promising in profits and energy organizations. In [269], it demonstrates through simulations the great potential presented by the south of Kazakhstan when it comes to implementing VPPs. This all confirms the great importance given to aggregators on the continent.

7.5. Regulatory and bureaucracy

Historically, most countries created their electrical networks and had public concessionaires that held the monopoly of generation, transmission, and distribution. Some countries still follow this more restrictive market organization, however, others believe that, by stimulating competition between different participants, there would be advantages in terms of price and quality for consumers, companies, and the government. This dilemma between market competition and public utilities is part of many studies due to its complexity, and balancing them to the point of acquiring the maximum possible advantages for everyone is one of the great challenges for policymakers, especially when it comes to subsidies and assistance [270].

This scenario is also becoming more complicated with the growth and dissemination of DERs [271,272], which are intrinsically decentralizing and lead to new questions and possibilities for the organization of electrical power systems. This has drawn significant attention over the last few decades, particularly regarding regulatory issues related to these technologies and prosumers [273].

They believe in general that countries with liberal electricity markets will prioritize decentralized generation while others with a more closed market will hinder the growth of DERs and, consequently, the formation of VPPs. Nevertheless, an unplanned legislative liberalization of the participation of DERs and aggregators in the wholesale market and the supply of AS can be harmful to society too, for this reason, the context of each country needs to be carefully analyzed.

In the text, [274], a comparison between two countries, South Africa and Germany, is made as an example of this issue. These two were chosen, because they both were previously dependent on carbon sources and began to invest in RES and decentralized regulations — Germany before and South Africa years later. However, how it was implemented in the African country ended up generating a series of problems, as it was not planned according to its context: there was no democratization of energy use, the state monopoly concessionaire barred a series of initiatives, there was little participation of national government on subsidies, etc. Germany also faced problems in challenging established configurations of socio-economic and political relations around the electrical system, but on a smaller scale. This demonstrates that the same measures, even if they seek to stimulate DERs and aggregators, can have negative effects depending on the country and its context.

The previous analysis is intrinsically related to the emergence and permanence of VPPs since they are part of the system's decentralizing solutions. It is worth noting, as seen in the article, that for these sets of DERs to emerge and establish themselves across the globe, a careful investigation of each electrical ecosystem must be made, identifying what are the proper measures to be adopted by policymakers. So that there is an environment of a healthy, productive, and advantageous market for society.

7.5.1. Market model 3.0

Recognizing the measures taken by one of these policies and legislation that did not remain static, but adapted to changes in the energy context, it is important to highlight the pioneering update of the Danish legislation called "Market Model 3.0" [275]. Before it occurs, the formation and permanence of aggregators in the Denmark electric ecosystem were very restrictive, as it still followed the traditional "top-down" structure in some points. Aggregators, for instance, had to be associated with a retailer and a Balance Responsible Party(BRP) to participate in the provision of AS. It was noted, however, that the main consequence of this type of obligatory dependency was a disincentive to the formation of the DERs sets by prosumers. Trying to change this limiting scenario and realizing that the increase in participation on the demand side could bring benefits to the network, measures were taken so that the legislation could adapt to the new context of decentralization.

Among those, the one related to pre-qualification in the provision of ancillary services stands out. It used to be made having as its main reference the conventional generators, which was an obstacle to the participation of VPPs mainly due to its stochastic nature. However, with updates of Market Model 3.0, it started to take into account the individualities of aggregators placing fewer outdated barriers to their participation- this does not mean arbitrary facilitation without criteria, but an update considering the new context that is developing in the country. Another change was related to the lesser dependence of aggregators concerning the BRP, because the obligatorily of a contractual connection, as it used to be done before any energy exchange, is no longer necessary- it is worth noting that there are about 50 BRPs in Denmark, which made difficult to implement DERs sets widely. Some liberalizations in the legislation of the Distribution System Operator (DO) monopolies, together with the stimulation of a healthy market ecosystem for the aggregators, proved to be also highly beneficial for the formation of these groups. With all these modifications, Denmark intends to achieve the "energy trilemma", making energy affordable, green, and with high security of supply.

7.5.2. Redispatch 3.0

Another regulatory change that introduces new measures to encourage the participation of decentralized energy resources is Redispatch 3.0 in Germany. In recent years, she has been encouraging an increase in RES over conventional power plants, which is positively affecting energy prices and reliability [276], however, it also increases the complexity of the system, especially if it is based on traditional structures.

Previously, the regulation was based on Redispatch 2.0, which was implemented in October 2015 as an amendment to the German Energy Act and is considered an important instrument for grid management in the energy transition to a more decentralized power system based on renewable sources [277]. The main feature of Redispatch 2.0 for RES is based on the fact that these resources have priority over conventional plants for compensation in case of curtailment, which is the need to reduce renewable generation due to grid congestion. This means that if a renewable energy source is reduced, it will receive compensation for lost revenue, which provides an incentive for renewable energy operators to participate in grid management. Additionally, Redispatch 2.0 encourages the deployment of energy storage systems and other flexible features that can help alleviate network congestion. However, it was noticed that this regulation could undergo updates in order to further encourage the integration of renewable sources and thus Redispatch 3.0 was created.

Redispatch 3.0 is a set of regulations introduced in 2021 to manage the increasing complexity of the electricity grid as the share of renewable energy sources is growing. One of the key benefits of Redispatch 3.0 is that it encourages the deployment and integration of DERs such as solar panels, wind turbines and energy storage systems. Furthermore, grid operators will be required to obtain flexibility from a variety of sources, including demand response, energy storage and flexible power generation, to balance the grid and avoid congestion, which is a regulatory update when compared to the last model. In addition, it promotes the use of digital technologies, such as real-time monitoring and control systems, to manage the network more efficiently and effectively. This is expected to improve the reliability and stability of the power grid, as well as facilitate the integration of more renewable energy sources by reducing the uncertainties inherent of them. In addition, while Redispatch 2.0 focused exclusively on high-voltage transmission networks, one of the main changes of the new model is the inclusion of low and medium-voltage distribution networks from an operator's point of view, which is responsible for the local distribution of electricity. By this extension on the regulation's approach, Redispatch 3.0 will enable greater integration of renewable energy sources at the local level and promote a more decentralized energy system in Germany.

8. Future perspectives

As seen throughout this paper, the decentralization of electrical networks is no longer just a futuristic promise, but a phenomenon that is becoming a reality across the globe. Through this, several forms of organization of distributed resources, such as energy generators or storage systems, are emerging. Among the main ones, P2P, LEMs, and VPPs deserve to be highlighted.

The first presents a series of challenges linked to essentially decentralized systems [64,65,72]. As seen, One of the main disadvantages of P2P systems compared to VPPs is their lower reliability due to their dependence on individual households and companies to manage their DERs. This can lead to a less stable network compared to VPPs that are mainly controlled by a single entity. Additionally, the decentralization of P2P systems can limit the flexibility in controlling loads and generators, making them less energy-efficient compared to VPPs [70]. Also, The lack of expertise in electricity management and optimization among individuals and companies using P2P systems can further reduce efficiency and optimization. Moreover, P2P systems can marginalize the role of DSOs and potentially reduce their profits, leading to reduced investment in grid improvement and outdated infrastructure. These issues can cause distribution problems, reduced process optimization, efficiency, and security of supply [69]. The Power Ledger project in Australia, a P2P energy trading platform, is an example of a project that faces difficulties due to these challenges. Local energy markets have the same disadvantages when compared to VPPs, however on a minor scale, mainly due to the lack of central coordination, which can lead to sub-optimal use of resources and grid instability. These analyses lead one to believe that VPPs are the most tangible options for organizing DERs in emerging decentralized systems.

It becomes evident that the implementation of a P2P system presents considerably greater challenges than VPPs. Conversely, LEMs can surpass some of these difficulties, particularly when adopting more coordinated approaches. As LEMs operate on a local scale, these issues can be mitigated and even overcome. Some remarkable studies have thoroughly analyzed the LEMs and shed light on their complexities [73,74]. It is worth mentioning that VPPs and LEMs can complement each other in specific scenarios. LEMs serve as a platform to promote decentralized energy production in specific regions, creating an environment where VPPs can thrive and engage in electricity transactions. On the other hand, the VPPs reinforce the presence of the LEMs, optimizing the distribution of resources in a given area and strengthening the negotiation capacity of the participants.

Because of that VPPs are increasingly becoming a reality in different parts of the world [7]. By aggregating and controlling distributed energy resources, they can provide a flexible and efficient way to balance electricity supply/demand and support grid stability. Hence, there are many perspectives on the potential of VPPs to transform the energy landscape and adapt to SGs [52,53]. Probably one of the biggest highlights of VPPs is their role in enabling the transition to a low-carbon energy system. As more renewable energy sources are integrated into the grid, the variability, and intermittency of generation become more difficult to manage, however, due to the organization and optimization of DERs, they can drastically reduce the need for fossil fuel generation [278]. Furthermore, aggregators help to unlock the potential of customer-owned DERs, which are becoming increasingly widespread as the costs of solar and battery storage continue to fall. By connecting these DERs through a VPP, electricity markets can access a large and diverse set of new resources that can aid, balance the grid and support the integration of renewable energies.

It can also be predicted that VPPs will have great integration with new technologies and resources [62]. Currently, the ones that have stood out the most and are being studied in this field are blockchain, AI, and IoT. The first can be used to allow secure and transparent peer-to-peer transactions between participants of electrical ecosystems. The second can use algorithms to optimize VPP dispatch and scheduling based on real-time data on weather, demand, and grid conditions. The third collect and communicate data from DERs, allowing for more efficient and effective management of VPPs and energy flow, reducing downtime, optimizing energy usage, and automating processes. Some examples of these technologies in practice are that smart technologies, electric vehicle chargers, and energy storage systems can be controlled remotely through the VPP platform to respond to changing grid conditions or to take advantage of dynamic pricing signals.

VPPs will continue to evolve in response to changing market conditions and regulatory frameworks around the world. As more utilities and energy markets adopt a decentralized approach to energy management, these aggregates are likely to become a key tool for optimizing the use of distributed energy resources and enabling the participation of prosumers in a greater market [59]. As seen, several jurisdictions around the world stand out for encouraging VPPs to participate in their energy market in the most diverse fields [275]. VPPs, for example, are already being used to participate in demand response programs and also provide grid services such as frequency regulation, voltage support, operational reserves, etc. that were traditionally managed by centralized power plants. And with the passage of time and the updating of legislation around the world about smart and decentralized electrical grids, VPPs will increasingly gain space as a great optimizer of distributed resources, especially renewable ones.

With more opportunities for participation, VPPs will excel in strengthening the network due to the provision of ancillary services [220,221]. Integrating multiple DERs will open up several new possibilities for network operations. Although recent, these groups already have a bright future in network operations. Studies and applications related to the integration of these systems with grids are already becoming a reality, and an increase in the integration of DERs, optimization of power systems, grid stability assistance, and continuous innovation of technologies allow VPPs to establish support and stability across multiple services [96,221]. Furthermore, VPPs will increasingly be used to support grid resilience against natural disasters [279], cyberattacks, and other disruptions. By featuring a range of varied energy resources, located in different locations and decentralized, VPPs can provide power that can help keep critical infrastructure and services online during emergencies. For example, cluster organizational control can be used to manage a network of distributed energy resources that can be quickly dispatched to replace damaged or destroyed network infrastructure, due to a disaster or failure.

Regarding electricity markets, aggregators also have enormous potential [55,56,58,59]. They will be a major player in these transactions, ultimately representing the one-way paradigm shift and allowing prosumers to fit into larger roles in the electrical environment than just passive consumers. In addition, increased competitiveness will be beneficial for buyers and resilience in general, as it increases the diversification of producers, also putting an end to the large energy monopolies that still prevail in some regions of the globe. Finally, the continuous and rapid innovations related to VPPs allows them to expand into new markets since renewable sources are fundamental in the energy transition from carbon sources, and VPPs have been proving to be efficient models to organize and optimize these features intermittently.

9. Conclusions

Typically, Distributed Energy Resources (DERs) have shown exponential and unplanned growth in recent years. This causes electrical instabilities and does not allow prosumers to provide and receive the full benefits of actively participating in the network. In trying to change this scenario, Virtual Power Plants (VPPs) have become an increasingly studied and implemented solution around the world. In addition to DER owners, it stands out for benefiting the Energy Market and System Operators through a series of models and strategies more suited to electrical decentralization. These aggregates also are formed, in general, by Renewable Energy Sources, which ends up meeting different environmental agendas.

The present work broadly brought several aspects of the aggregators described in the current literature. It began by analyzing the various existing categories and also presenting the representative theoretical and mathematical details of each of them. From the analyses carried out, it is clear how recent the theme is, due to the wide variation of classes with different parameters and perspectives, i.e., there is no single and universal classification, but models that fit better according to the research of different authors.

After that, the necessary technologies for achieving the objectives of a VPP were described, which, usually, are associated with data processing and the exchange of energy, information, and money. As it was seen, VPPs establish themselves as major organizers of resources in the next years, these components must maintain the downward trend in their prices. Along with this, new studies related to the useful life of devices, like batteries or photovoltaic panels, will lead to greater coverage of DERs as a worthwhile investment for people. Regarding the most advanced technologies, a set of factors can be consolidated and improved such as data collection, result prediction, reinforcement learning, optimization algorithms, Artificial Intelligence, etc. All these improvements are necessary to transform theoretical concepts of VPPs, often complex, into practical and tangible applications beyond academic circles.

Another important point is related to the relationships that will emerge in the electrical networks with the entry of the VPP as a new participant. Several authors predict that a dispute of interests will form and try to find methods to solve them through models proposed by game theory. In general, the government will play a regulatory role in them, keeping the objectives of all participants balanced and considering the context of the country. These studies are extremely important for the transition from an essentially centralized network to a decentralized one, since if they are not done correctly, a hostile environment can be formed both for VPPs, making their permanence and implementation difficult, and for traditional entities, endangering their profits and survival.

In addition to this macro organization, it was also revised the internal arrangement of the VPPs. They can use not only the methods proposed by game theory, but also more centralized models, as in Fog as a VPP, or decentralized ones, as in Peer-to-Peer or distributed algorithms. The best system must be chosen according to the objectives and equipment of each VPP. Currently, the following models of Organizational control stand out: Centralized control, decentralized control, distributed control, and Comprehensive control. Within this internal model, it is worth mentioning the importance of acquiring an adequate and advantageous tariff system for the different categories of prosumers, bringing benefits and encouraging them to participate more actively in electrical power systems, since, if it is not economically advantageous to participate in these arrangements, they will choose to proceed alone — reducing the emergence of VPPs.

Possibly one of the greatest advantages of these DER clusters lies in their ability to provide Ancillary Services. Despite the existence of research and articles analyzing this topic, there is still a long way to go before reaching a concrete conclusion on the extent to which the VPPs can participate in these aids — this uncertainty also comes from the few cases implemented nowadays. From the published studies, it can be seen that the categories of “Frequency services” and “Reactive power and voltage control” are the ones that are currently best detailed, mainly due to the great capacity of DERs to quickly deal with variations in their expected pattern. On the other hand, “Black-start services”, “Scheduling and dispatch” and “Operating reserves” have fewer publications, which reveals a potential area for future studies. Even so, it is important to emphasize that all AS must be much more explored, since the full potential of the VPPs is not known and, because, they will be a key point in the permanence of these new virtual institutions by helping the entire electric power system, that has been experiencing difficulties with the scenario of decentralization and big electrification.

Although all the aforementioned points are fundamental for VPPs, they will not be useful in practice if regulatory and policy issues are not taken into account. As seen in Section 7, the dilemma between greater liberalization of demand-side and bigger control by public utilities represents a major dilemma for policymakers. Government officials play a fundamental role in this process by choosing which will be the best direction for their countries’ energy policy and where greater investments will be made. However, if these paths taken are not planned according to the individual context of each country, a large implementation of VPPs will likely fail to fulfill its objectives, i.e, not leading to big energy democratization, adequate service provision, or efficient financial economy. Given this and the analyses made in the respective section, it is clear that there is not a single universal path that will work in all countries, but different paths that will lead to more or less satisfactory results according to each society. Some examples of countries that have more coherent positions according to their individualities were addressed by this review highlighting de Denmark’s case with Market Model 3.0 and German’s with Redispatch 3.0.

In conclusion, the review presented not only consolidates existing knowledge on VPPs but also identifies key areas for future research and development. By addressing both the theoretical and practical aspects of VPPs, this work aims to provide a holistic understanding that can guide policymakers, researchers, and industry stakeholders in the effective implementation and optimization of VPPs. The insights gained from this review underscore the importance of continued innovation, supportive regulatory frameworks, and strategic investments to unlock the full potential of VPPs, ultimately contributing to a more resilient, efficient, and sustainable energy future.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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